
LADDER: LLM Agent Diagnostic of Domain-level Estimation Reliability

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Abstract

1 Users learn where each LLM agent is reliable; the agent itself does not. We ask
2 a single mechanistic question: at which sub-capability does this self-modeling
3 break down? Three candidates—H1 output channel, H2 prior metacognition,
4 H3 noise-robust online inference—are conflated by standard benchmarks. We
5 build **LADDER**, a diagnostic benchmark whose five-level oracle ladder (L0 raw
6 confidence $\rightarrow$ L4 full-statistic oracle) lets each adjacent comparison rule out one
7 candidate, scored with **Capability Boundary Fidelity (CBF)**, a per-domain metric
8 distinct from per-query calibration (ECE). The ladder yields a clean localization
9 across 19 models. H1 is ruled out (GT-SelfStats CBF ≥ 0.99 on every model);
10 H2 is partial and family-split rather than the bottleneck (GPT models reach 0.795–
11 0.859 zero-shot, while the three Claude-4-6/4-7 models fall below NoMemory). H3
12 is the locus: first-order single-rater learners cluster near NoMemory, while explicit
13 *second-order* estimators (asymmetric Dawid–Skene, GLAD) substantially exceed
14 the L1 prior, with GLAD winning on all nine frontier models (binomial sign-test
15 $p = 0.002$). The pattern replicates across task families (13/19 sign-flip rate on
16 extension tasks), implying that per-task boundary modeling is also needed. We
17 name this failure mode **Self-Boundary Collapse** and identify noise-robust second-
18 order online inference as the central remaining challenge. We release the ladder,
19 CBF, cross-family user simulator, and trace dataset as benchmark infrastructure.

20 1 Introduction

21 LLM agents are increasingly used in long-horizon, multi-turn settings, including customer service,
22 coding, research, and tool use. Over time, users develop an empirical sense of where an agent is
23 reliable. The agent, however, typically retains no comparable self-model: each session begins without
24 prior calibration and ends by discarding its own evidence. This asymmetry is well documented, but
25 the underlying *capability gap* remains poorly localized. In particular, we do not yet know *where*,
26 mechanistically, agent self-modeling fails.

27 We distinguish three possible bottlenecks. Under H1, the output channel is the limiting component:
28 the model cannot format and emit numerical per-domain statistics. Under H2, the model lacks an
29 accurate prior over its own per-domain abilities, so any useful self-assessment must be *learned*
30 through interaction. Under H3, the model can state reasonable priors but cannot update them
31 reliably from *noisy* implicit user feedback. These hypotheses imply different intervention targets.
32 H1 suggests fine-tuning for numerical reporting; H2 points to pre-training-time metacognitive
33 curricula; H3 calls for noise-robust online estimation. Existing work on agent self-knowledge [3, 16],
34 calibration [13, 18, 44], and memory systems [25, 32, 34, 39, 46] measures important symptoms but
35 does not isolate which of these bottlenecks is responsible. We refer to the resulting failure mode,
36 in which LLM agents repeatedly receive feedback yet fail to convert it into a stable domain-level

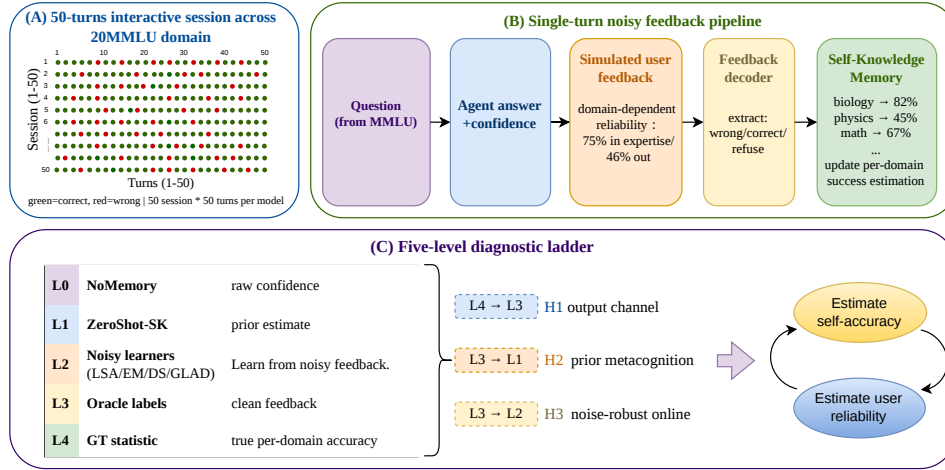


Figure 1: **The L0–L4 oracle ladder localizes where agent self-knowledge fails.** (A) 50 sessions $\times$ 50 turns over 20 MMLU domains (green=correct, red=wrong). (B) Single turn: agent answers with confidence; a cross-family simulated user gives noisy feedback ($r_{in} \approx 75\%$, $r_{out} \approx 46\%$); a decoder extracts the label that updates the per-domain self-model. (C) The five diagnostic levels with adjacent comparisons isolating H1 (output channel, ruled out at L4–L3), H2 (prior metacognition, family-split at L3–L1), and H3 (noise-robust online inference, the locus at L3–L2).

self-model, as **Self-Boundary Collapse**. Our central question is where this failure appears on the H1/H2/H3 ladder.

We study this question with LADDER. The benchmark is organized around an oracle-bracketed *five-level baseline ladder*: L0 raw confidence (NoMemory), L1 zero-shot per-domain prior, L2 noisy-feedback learners (LLM-self-assessment, classical recalibration, EM-joint, second-order Dawid–Skene/GLAD), L3 binary-label oracle, and L4 full-statistic oracle. Adjacent comparisons localize the failure. L4–L3 tests the output channel (H1), L3–L1 tests the prior (H2), and L3–L2 tests online inference from noisy feedback (H3). The protocol pairs each agent with a domain-heterogeneous simulated user, reliable 75% of the time in-expertise and 46% out, with cross-family routing to mitigate a dual-role bias that we quantify in §4.3. Sessions last 50 turns, and the resulting per-domain self-model is evaluated with **Capability Boundary Fidelity (CBF)**, a metric we introduce to complement per-query calibration measures such as ECE.

Across 19 models from 8 families and 60K+ interactions, the ladder gives a clear localization. H1 is ruled out: GT-SelfStats reaches $CBF \geq 0.99$ on every model, showing that the output channel can faithfully emit per-domain statistics when supplied. H2 is real but not the main bottleneck. The zero-shot prior is the strongest non-oracle baseline for GPT models, with CBF 0.795 on GPT-5.4 and 0.859 on GPT-5.5, but it falls below NoMemory on all three Claude-4-6/4-7 models, 0.460–0.482 vs. 0.557–0.603; the frontier mean ($n=9$) is 0.654. Thus prior self-knowledge exists in some families and is absent in others, but it never reaches the oracle ceiling. The main failure is H3. First-order single-rater learners, including UF-PDC, Platt, HistBin, Bayes, and simple EM-Joint, cluster near NoMemory on the frontier ($n=9$ CBF 0.623–0.625), and cannot recover per-domain accuracy from noisy feedback. Explicit *second-order* estimators do substantially better: asymmetric Dawid–Skene reaches a frontier mean of 0.768, and GLAD reaches 0.742. GLAD beats L1 on all nine frontier models, with binomial sign-test $p = 0.002$. A remaining L2→L3 gap of roughly 0.23 CBF on the frontier (best L2 DS-asym 0.768 vs. OraclePDC 1.000) persists, but it appears tractable rather than fundamental. Figure 1 summarizes the protocol, the noisy-feedback channel, and the diagnostic localization in three panels.

Contributions. (i) A benchmark whose adjacent-level comparisons isolate one sub-capability each, supported by an oracle-bracketed five-baseline ladder; (ii) the per-domain Capability Boundary Fidelity (CBF) metric, distinct from per-query ECE; and (iii) empirical evidence—across 19 models

from 8 families and 60K+ noisy-feedback turns—that an H3-class noise-robust online estimator is the central remaining bottleneck for agent self-modeling, with explicit second-order estimators (DS-asym, GLAD) the most promising direction.

2 Related Work

LADDER sits at the intersection of four lines of work—single-shot LLM self-knowledge and calibration benchmarks, multi-step agent overconfidence under clean feedback, multi-turn agent benchmarks that score task completion rather than self-knowledge, and inference-time memory and feedback-learning systems. We position LADDER against each line and summarize the design-axis comparison in Table 1; the diagnostic distinction is that LADDER is the only benchmark that combines a noisy domain-dependent feedback channel with an oracle-bracketed accumulation horizon.

LLM self-knowledge and calibration benchmarks. A growing line of work probes whether LLMs can represent or report their own correctness on a single query. Internal-representation studies show that latent confidence signals exist [4, 17, 18], while recent benchmarks evaluate metacognition behaviorally and find that frontier models exhibit only partial, context-dependent self-monitoring [1, 29]. Table 1 positions the most closely related benchmarks against LADDER along four design axes: feedback channel, horizon, whether self-knowledge is accumulated, and whether the protocol forces second-order user-reliability inference.

Table 1: **Related-benchmark positioning along four design axes.** LADDER is the only benchmark combining a noisy interactive feedback channel, accumulation horizon, and second-order user-reliability inference.

Benchmark	Feedback	Horizon	Self-knowledge	2nd-order
SAD [22], SelfAware [50]	none	single-shot	static probe	—
Barkan [3]	clean step signals	multi-step exec	per-task	—
MemoryAgentBench [16]	implicit (mem ops)	long-horizon	exogenous mem only	—
AgentBoard [30], τ -bench [48]	task reward only	multi-turn	none	—
Knowledge-boundary [24, 41]	none	single-shot	static	—
LADDER (ours)	noisy NL, 75/46 asym	50-turn session	per-domain accumulated	✓

Single-shot self-knowledge probing. SAD [22] and SelfAware [50] score self-knowledge per query without temporal dynamics; their findings supply the “L1 prior” anchor of our ladder. LADDER uses the L1→L3 contrast to ask whether *noisy interaction* improves on that prior—finding it does not.

Multi-step agent overconfidence. Barkan et al. [3] measure whether 13 frontier LLMs can predict their own success on BigCodeBench and SWE-Bench under multi-step tool execution and find all models overconfident with overconfidence *worsening* during execution. Their setting gives clean per-step success/failure signals; ours gives noisy domain-dependent natural-language feedback, where first-order methods cluster at the L2 floor. Barkan establishes that overconfidence persists under clean traces; we localize *where* the failure lives when feedback itself is noisy, via the L0–L4 oracle-bracketing ladder.

Multi-turn agent benchmarks without self-knowledge. AgentBoard [30] and τ -bench [48] evaluate task completion under multi-turn interaction but do not measure self-knowledge; their reward signal is task success, not self-estimate accuracy. LADDER holds the underlying task simple (MCQ) and varies only the information-access regime, so the diagnostic ladder isolates self-knowledge mechanisms these benchmarks cannot disentangle from task-execution confounds.

Static knowledge-boundary benchmarks [12, 15, 24, 26, 27, 41, 49] share the same structural limitation: they evaluate *a priori* boundary awareness, not whether the agent’s self-model can be accumulated under noisy interactive turns (Appendix G.2).

Agent memory and learning from implicit feedback. Existing memory systems—MemGPT [34], Mem0 [32], RMM [39], A-MEM [46], MemOS [25], MEM1 [54]—and the recent MemoryAgentBench [16] all store *exogenous* state (user facts, dialogue, trajectories). None model the agent’s *own*

per-domain reliability; we introduce this as a fifth memory competency. On the feedback-learning side, RLHF [33] and DPO [36] update weights at training time assuming reliable annotators; Liu et al. [28] establish that implicit dialogue feedback is noisy and only partially useful as a distillation signal; Leng et al. [23] show RLHF itself induces systematic overconfidence. LADDER studies the orthogonal regime: *inference-time* accumulation of *noisy, domain-dependent* feedback into a per-domain self-model, with no parameter updates. The unique twist is asymmetric reliability (75% in-expertise vs. 46% out), which turns boundary learning into a *second-order* problem.

Selective prediction and task-specific calibration. Per-query selective-prediction methods [2, 19, 20, 40, 42, 45, 52] treat each prediction in isolation. LADDER instead enables *historical* selective prediction: abstention based on persistent per-domain self-model statistics. Independently, prior work has noted that calibration does not transfer across tasks [6, 31, 37]; our experiments quantify this systematically across MMLU, GSM8K, HumanEval, and BFCL. The closest concurrent benchmarks [5, 10, 11, 14, 53] either assume perfect feedback, require weight updates, or focus on single-agent dynamics; none combine multi-task evaluation, noisy domain-dependent feedback, and a quantitative boundary-fidelity metric over extended horizons.

3 LADDER

LADDER evaluates a single capability: given a stream of (q_t, a_t, c_t, f_t) tuples over T turns—a question q_t from a hidden domain d_t , the agent’s answer a_t with confidence $c_t \in [0, 1]$, and a natural-language user feedback f_t —can the agent build a function $\hat{p} : \mathcal{D} \rightarrow [0, 1]$ that closely approximates its true per-domain accuracy $p^*(d)$?

The benchmark instantiates this question via a 50×50 protocol: 50 sessions $\times$ 50 turns per session, totaling 2,500 turns per model, drawing from 20 MMLU subjects [15] partitioned per model into Strong/Medium/Weak tiers relative to that model’s own accuracy distribution. The user is simulated by a cross-family LLM (avoiding the dual-role bias of Appendix E.5), parameterised by an expertise set $\mathcal{E} \subset \mathcal{D}$ with empirically calibrated reliabilities of $\sim 75\%$ in-expertise and $\sim 46\%$ out.

We additionally construct three extension tasks—GSM8K [8], HumanEval [7], and BFCL tool-calling [47], collectively called the *extension suite*—to test cross-task generalisation (Appendix F.1). Full task formalisation, session protocol, question-pool partitioning, and persona parameterisation are deferred to Appendix B.1.

Capability Boundary Fidelity (CBF). Our central metric scores how closely the agent’s self-assessed per-domain accuracy tracks the truth:

$$\text{CBF} = 1 - \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} |\hat{p}(d) - p^*(d)|, \quad \text{CBF} \in [0, 1]. \quad (1)$$

CBF is computed once per session at the end of the 50-turn protocol. CBF is complementary to per-query calibration metrics (ECE/Brier): ECE operates on (c_t, y_t) pairs while CBF operates on $(\hat{p}(d), p^*(d))$ pairs, so neither implies the other; Figure 2 confirms this empirically (Pearson $r = -0.10$, $p = 0.67$ across all 19 models). The trivial estimator $\hat{p}(d) = 0.5$ attains $\text{CBF} = 1 - \mathbb{E}_d[|0.5 - p^*(d)|]$, which is high for weak models clustered near 50% and low for strong models with widely varying accuracy across domains; this asymmetry is exactly what makes the metric informative for our diagnostic.

Baseline-adjusted CBF: the primary cross-model metric. To remove the trivial-estimator artifact when comparing methods *across* models, we define a normalized variant that measures the fraction of the oracle gap a method closes relative to the constant baseline:

$$\widetilde{\text{CBF}}(m, \text{method}) = \frac{\text{CBF}_{\text{method}}(m) - \text{CBF}_{\text{NoMem}}(m)}{\text{CBF}_{\text{Oracle}}(m) - \text{CBF}_{\text{NoMem}}(m)}, \quad (2)$$

where $\text{CBF}_{\text{NoMem}}(m)$ uses the model’s verbalized confidence (which empirically clusters near 0.5 on the trivial domains) as the constant-baseline anchor, and $\text{CBF}_{\text{Oracle}}(m) \geq 0.99$ is the L3 OraclePDC ceiling. Values are interpretable as “fraction of L0→L3 gap closed”: $\widetilde{\text{CBF}} = 1$ matches oracle, 0 ties the trivial baseline, < 0 underperforms it. We report CBF as the primary metric for cross-model means and ladder-headline comparisons (§4.2), and use raw CBF only as a within-model diagnostic.

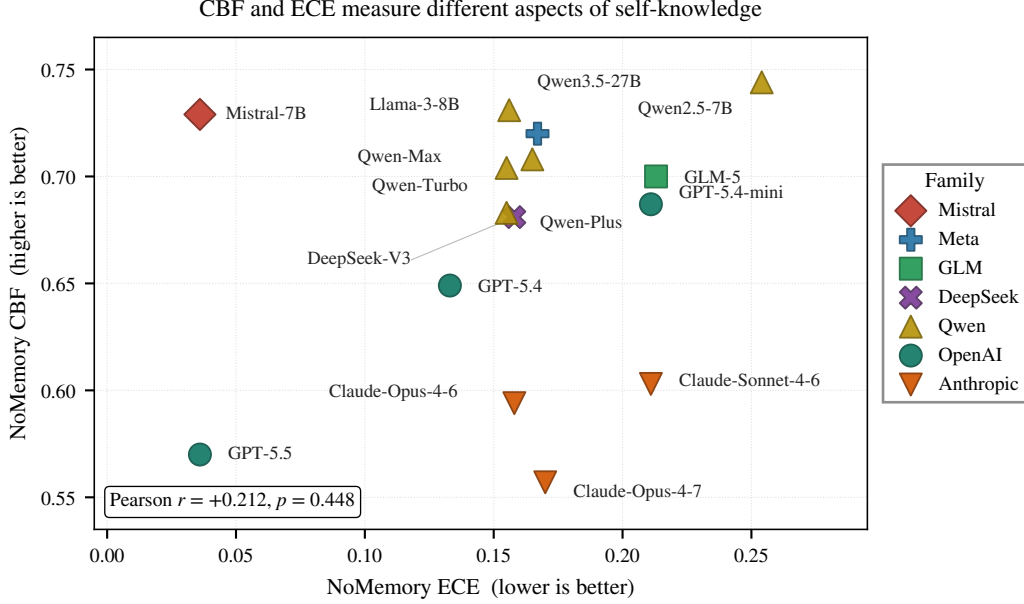


Figure 2: **CBF and ECE measure different aspects of self-knowledge.** Per-model NoMemory CBF vs. NoMemory ECE across 19 models; Pearson $r = -0.10$ ($p=0.67$) confirms the two metrics are essentially uncorrelated.

Auxiliary metrics and domain structure. We additionally report Hallucination Rate, ECE, and Brier as auxiliary calibration metrics; their definitions follow Guo et al. [13] and are given in Appendix B.1. CBF requires a multi-domain task: for single-domain extension tasks (GSM8K, HumanEval, BFCL) we report only ECE and confidence–accuracy gap, and the cross-task analysis in Appendix F.1 uses these standard metrics.

Cross-model comparability. The 20 MMLU subjects are partitioned into Strong/Medium/Weak tiers *relative to each model’s own held-out validation accuracy* so that every model faces a non-trivial cross-domain spread. CBF is therefore a within-model diagnostic; cross-model means reported throughout summarize per-model values that were each computed against the model’s own $p^*(d)$, and are appropriate for comparing *methods within the diagnostic ladder*, not for ranking models on absolute capability (Appendix B.1).

Baselines: a five-level diagnostic ladder. We evaluate ten baselines organized as a ladder of *information access*, designed so that comparing adjacent levels isolates a specific sub-capability:

- **L0 (raw): NoMemory**—verbalized confidence with no memory, the stateless anchor.
- **L1 (prior only): ZS-SK** (ZeroShot Self-Knowledge)—the agent is asked, with no session history and no feedback, to estimate its own per-domain accuracy directly; this measures *prior metacognition*.
- **L2 (learning from noisy feedback): LSA** (LLMSelfAssess; in-context reasoning over session history), *UF-PDC* (LLM-decoded per-domain counting), *Platt-Online* [35], *HistBin-Online* [51], *Bayes-Domain* (Beta posterior per domain), and *EM-Joint* (joint estimation of $\hat{p}(d)$ and $\hat{r}(d)$). These differ in their inference machinery but all consume the same noisy feedback stream.
- **L3 (binary-label oracle): OraclePDC**—per-turn ground-truth correctness labels, with the agent computing per-domain running averages. Tests whether the agent could build a self-model *if feedback were clean*.
- **L4 (format-channel sanity ceiling): GT-SelfStats**—the agent is given $p^*(d)$ directly in the prompt and asked to emit them in the structured response format. This is a sanity check that the L2 gap cannot be attributed to an inability to format and emit per-domain statistics. We do not claim it tests verbalization-of-self-knowledge.

Table 2: **Main results: end-of-session CBF across 19 models on the five-level baseline ladder** (50 sessions $\times$ 50 turns; higher better; bold = best non-oracle per row).

Model (Acc)	L0	L1	L2 (learn from noisy feedback)							Oracles	
	NoMem	ZS-SK	LSA	UF-PDC	Platt	HistBin	Bayes	EM	Best	OraPDC	GT-Stats
Mistral-7B (42%)	0.729	0.729	0.729	0.720	0.717	0.717	0.731	0.732	EM	1.000	1.000
Llama-3-8B (55%)	0.720	0.711	0.549	0.715	0.724	0.724	0.730	0.731	EM	1.000	1.000
GLM-5 (55%)	0.700	0.700	0.700	0.689	0.696	0.696	0.711	0.714	EM	1.000	1.000
DeepSeek-V3 (62%)	0.681	0.673	0.721	0.681	0.683	0.683	0.696	0.699	LSA	1.000	1.000
Qwen-Turbo (64%)	0.708	0.730	0.703	0.697	0.712	0.712	0.724	0.728	ZS	1.000	1.000
Qwen2.5-7B (65%)	0.744	0.663	0.701	0.738	0.745	0.745	0.755	0.757	EM	1.000	1.000
Qwen-Max (71%)	0.704	0.766	0.781	0.660	0.684	0.684	0.702	0.705	LSA	1.000	1.000
Qwen-Plus (79%)	0.683	0.735	0.435	0.686	0.695	0.695	0.700	0.701	ZS	1.000	1.000
Qwen3.5-27B (79%)	0.731	0.731	0.731	0.712	0.720	0.720	0.740	0.746	EM	1.000	1.000
GLM-5.1 (76%)	0.600	0.600	0.600	0.605	0.609	0.609	0.613	0.615	EM	1.000	1.000
<i>Frontier models:</i>											
GPT-5.4-mini (72%)	0.687	0.769	0.763	0.687	0.701	0.701	0.706	0.706	ZS	1.000	0.995
GPT-5.4 (82%)	0.649	0.795	0.781	0.668	0.675	0.675	0.674	0.677	ZS	1.000	0.994
GPT-5.5 (94%)	0.570	0.859	0.902	0.601	0.612	0.612	0.603	0.605	LSA	1.000	0.992
Claude-Opus-4-7 (69%)	0.557	0.482	0.641	0.593	0.577	0.577	0.571	0.570	LSA	1.000	0.992
Claude-Sonnet-4-6 (86%)	0.603	0.472	0.720	0.640	0.617	0.617	0.613	0.609	LSA	1.000	0.993
Claude-Opus-4-6 (90%)	0.594	0.460	0.680	0.618	0.597	0.597	0.603	0.604	LSA	1.000	0.993
DeepSeek-V4-Pro (90%)	0.575	0.849	0.675	0.593	0.609	0.609	0.602	0.605	ZS	1.000	0.992
Qwen3.6-Max (92%)	0.580	0.580	0.580	0.586	0.599	0.599	0.598	0.601	EM	1.000	1.000
Qwen3.6-Plus (93%)	0.617	0.617	0.617	0.626	0.636	0.636	0.638	0.642	EM	1.000	1.000
<i>Within-tier means:</i>											
Frontier (n=9; all columns)	0.604	0.654	0.707	0.623	0.625	0.625	0.623	0.624	LSA	1.000	0.993
Strong-mid (n=4)	0.680	0.708	0.637	0.665	0.677	0.677	0.689	0.692	ZS	1.000	1.000
Mid (n=3)	0.711	0.689	0.708	0.705	0.713	0.713	0.725	0.728	EM	1.000	1.000
Weak (n=3)	0.716	0.713	0.659	0.708	0.712	0.712	0.724	0.726	EM	1.000	1.000

180 All feedback decoding (L2 + UF-PDC) uses an LLM-based decoder rather than rule-based pattern
181 matching, more aligned with the “self-knowledge boundary” setting and avoiding rule-classifier
182 coverage as a confound. Full per-baseline algorithms are in Appendix C.1.

183 4 Diagnostic Results: Where Does Agent Self-Knowledge Fail?

184 We test the L0–L4 ladder on 19 models across seven families and 60K+ interactions. §4.1 fixes
185 the protocol, model set, and decoder. §4.2 reports per-model end-of-session CBF along the ladder,
186 separating the $n=9$ frontier subset from the rest. §4.3 reads the ladder vertically by adjacent-level
187 gaps to localize the failure. §4.4 replicates the headline pattern on the GSM8K, HumanEval, and
188 BFCL extension tasks under a protocol-controlled subset analysis. The headline: H1 is ruled out, H2
189 is family-split, H3 is the locus.

190 4.1 Setup

191 We evaluate nineteen models from eight families spanning 42–94% MMLU accuracy (Appendix G.3).
192 Each model–baseline pair runs 2,500 turns (50 sessions $\times$ 50 turns); user personas are simulated by
193 a cross-family LLM with three expertise domains¹; answer extraction uses a lenient multi-strategy
194 parser tolerant of free-form responses to avoid truncating reasoning models. Feedback decoding for
195 all L2 baselines uses an LLM-based decoder. Open-weights models (Mistral-7B, Llama-3-8B) are
196 served via vLLM from pinned snapshots and serve as the reproducibility anchor; API access dates,
197 simulator-routing rules, and the parser specification are deferred to Appendix G.3.

4.2 Main Results

Three diagnostic patterns dominate Table 2: (i) algorithmic L2 methods collapse to NoMem on every frontier model; (ii) the zero-shot prior splits cleanly by family, with GPT estimates well-calibrated and Claude estimates inflated; (iii) only second-order estimators (DS-asym, GLAD) and in-context LSA exceed the L1 prior on the frontier. We examine each pattern below.

Algorithmic L2 collapses on frontier models; ZS and LSA do not. For the two highest-accuracy frontier models, the five *algorithmic* L2 baselines (UF-PDC, Platt, HistBin, Bayes, EM-Joint) cluster near NoMem (GPT-5.5: 0.601–0.612, all ~ 0.03 – 0.04 above NoMem 0.570; Claude-Opus-4-7: 0.570–0.593, all ~ 0.01 – 0.04 above NoMem 0.557). Both LSA and the L1 prior break this collapse decisively on GPT-5.5 (LSA 0.902, ZS-SK 0.859: LSA is $+0.290$ over best algorithmic L2 0.612, ZS is $+0.247$ over the same) and on Claude-Opus-4-7 (LSA 0.641 vs. NoMem 0.557, $+0.084$). The cross-frontier pattern: algorithmic L2 methods fail to surpass NoMem on every frontier agent; LSA strictly exceeds NoMem on 7/9 frontier rows (all 3 GPT, all 3 Claude, DeepSeek-V4-Pro) and ties on the two Qwen3.6 rows; ZS strictly exceeds NoMem on 4/9 (all 3 GPT plus DeepSeek-V4-Pro), ties on the two Qwen3.6 rows, and falls below NoMem on all 3 Claude rows. In particular, GPT-5.5 is now decisively dominated by LSA (0.902, the benchmark’s highest non-oracle CBF) followed by ZS (0.859); both substantially exceed every algorithmic L2 method (≤ 0.612), and the earlier 10×50 result in which LSA fell well below NoMem (0.623) does not reproduce at the larger session budget. Held-out evaluation and second-order estimators (Dawid–Skene, GLAD) are reported on the full $n=9$ frontier; the MMLU-Pro and BBH contamination controls (Appendix E.1) cover the full $n=9$ frontier set.

H1 ruled out as the limiting factor ($L4 \approx 1.0$). GT-SelfStats reaches $CBF \geq 0.99$ on every one of the 19 models (worst gap to perfection 0.008 on Claude-Sonnet-4-6 and DeepSeek-V4-Pro). When the true per-domain statistics are placed in the prompt, every tested model can format and emit them: the L2–L1 gap is not located in the output channel.

H2 splits cleanly by model family: GPT well-calibrated, Claude inflated. Across the 9 frontier models, LSA averages CBF 0.707 vs. ZS-SK’s 0.654 and the best algorithmic L2 method’s 0.625 ($+0.053$ LSA-over-ZS margin). LSA wins on the three Claude models and on GPT-5.5 (LSA 0.902, the benchmark’s highest non-oracle value); ZS wins on GPT-5.4 (0.795) and GPT-5.4-mini (0.769). The split is driven by Claude rows where ZS is systematically inflated and falls below both NoMem and the algorithmic L2 cluster. On the strong-mid tier ($n=4$) ZS leads on average (0.708 vs. best L2 EM-Joint 0.692); for mid and weak tiers, EM-Joint ties modestly above the prior, and the trivial $\hat{p} \approx 0.5$ anchor remains competitive. Per-model breakdowns and the LSA catastrophic-failure case on Qwen-Plus appear in Appendix E.1.

H3 localized: first-order online learning fails; second-order estimators and in-context aggregation succeed. The five first-order L2 methods (UF-PDC, Platt-Online, HistBin-Online, Bayes-Domain, EM-Joint) cluster in 0.62–0.63 mean CBF on the frontier ($n=9$) and fail to exceed L1 ZS (0.654). Three method classes break this floor: GLAD reaches 0.742 with **9/9 frontier wins** (binomial $p = 0.002$, per-model deltas $+0.028$ to $+0.240$, bootstrap 95% CI $[+0.060, +0.135]$); DS-asym reaches 0.768 with 8/9 wins ($p = 0.020$); LSA reaches 0.707 with 4/9 wins, exceeding L1 on Claude (3/3) but inconsistently elsewhere. The H3 framing thus holds for first-order single-rater estimators only; second-order estimators and in-context aggregation substantially exceed L1, yet none reaches L3—the residual L2-vs-L3 gap is ≈ 0.23 on the frontier (DS-asym 0.768 vs. OraclePDC 1.000) and ≈ 0.27 on the cross-model average ($n=19$). On non-frontier tiers, EM-Joint leads modestly (mid $+0.012$, weak $+0.006$), winning outright on Llama-3-8B and Qwen2.5-7B (Appendix E.1 for full breakdown).

Best-baseline assignment. The argmax over the 8 non-oracle columns of Table 2 splits three ways across 19 models: **LSA** wins on 6 (DeepSeek-V3, Qwen-Max, GPT-5.5, all 3 Claude); **EM-Joint** on 8 (mostly weak/mid, plus the 3 new Qwen3.6/GLM-5.1 rows where the EM–NoMem–LSA

¹Frontier set: 9 closed-source models from 4 families (GPT, Claude, DeepSeek, Aliyun), used for within-tier means and DS/GLAD analysis. MMLU-Pro and BBH controls (Appendix E.1) cover the full $n=9$ frontier set.

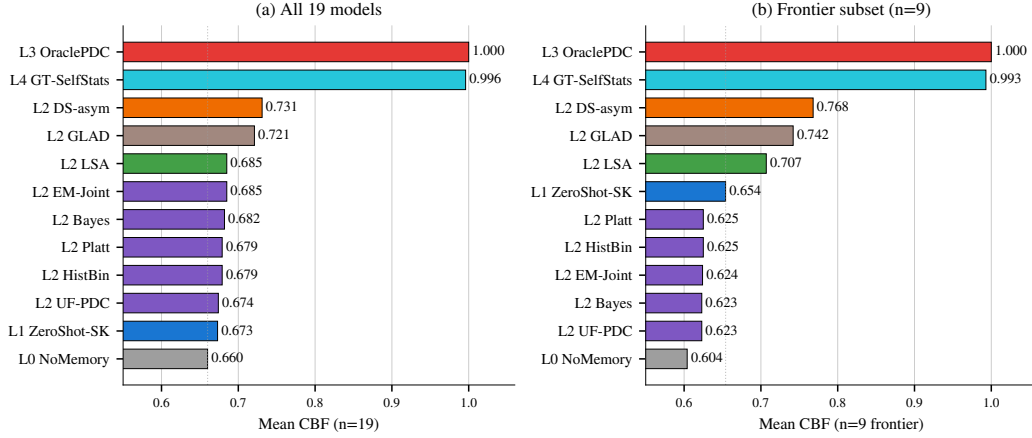


Figure 3: **The L0–L4 ladder localizes the bottleneck.** Mean CBF per baseline level on (a) all 19 models and (b) the 9 frontier models; the L2→L3 gap of ≈ 0.27 is the empirical bottleneck. Dotted vertical line marks the L0 NoMemory anchor.

spread is within sampling noise); **ZS-SK** on 5 (Qwen-Turbo, Qwen-Plus, GPT-5.4, GPT-5.4-mini, DeepSeek-V4-Pro). NoMemory is never the unique best.

4.3 Diagnosis via the Ladder Visualization

Figure 3 summarizes the per-model results of Table 2 as a vertical ladder in which each adjacent gap functions as a sub-capability test: the bottleneck is not the output channel ($L4 \approx L3 \approx 1.0$) but noise-robust online inference ($L2 \rightarrow L3$ gap ≈ 0.27).

Reading the ladder by adjacent comparisons: $L4$ vs $L3$ ($< 1\%$ gap)—the output channel is not the constraint, ruling out H1. $L3$ vs $L2$ (~ 0.27 gap)—the empirical bottleneck: with clean per-turn labels agents reach 1.0, but no method fully recovers the signal from noisy LLM-decoded feedback, confirming H3. $L2$ vs $L1$ ($n=19$: ≈ 0.67 algorithmic L2 vs. 0.673 ZS-SK; $n=9$ frontier: 0.623–0.625 vs. 0.654)—L1 ZS ties or exceeds the algorithmic L2 cluster on average; only second-order estimators (DS-asy, GLAD) and in-context LSA exceed L1 on the frontier (Appendix E.1, A5 and A6).

A diagnostic per-model selection rule (R4). The H2 family-split above motivates a simple diagnostic rule rather than a new estimator: **R4** (per-model best-of-two between the prior and in-context aggregation) selects L1 (ZS-SK) if $ZS(m) > NoMem(m)$, otherwise L2 (LSA). The “R4” label is internal to our four-rule analysis pipeline (R1–R4 documented in Appendix E.7); we report only R4 here because the R1–R3 rules are subsumed. R4 attains mean CBF ≈ 0.71 across $n=19$ and closes a substantial fraction of the always-ZS-to-per-model-oracle gap, but its per-model sign tests against always-ZS and always-LSA are non-significant: the rule *redistributes* wins, it does not dominate. R4 serves as a benchmark-validation lower bound: any new method should clear it. The empirical frontier ceiling rises to DS-asy at CBF 0.768 in the $n=9$ extension. Full LOOCV, per-model breakdown, and sign-test arithmetic in Appendix E.7.

4.4 Cross-Task Heterogeneity: Per-Task Boundary Models Are Necessary

Across MMLU, GSM8K, HumanEval, and BFCL the same model exhibits qualitatively different calibration regimes. We measure the per-task confidence–accuracy gap (mean verbalized confidence minus mean accuracy) and define a *sign-flip* as a model whose gap changes sign across two tasks: a model overconfident on one task (gap > 0) but underconfident on another (gap < 0). Within the protocol-controlled extension suite (GSM8K/HumanEval/BFCL under the same single-shot two-step protocol), 13 of 19 models show at least one such sign-flip across the three same-protocol tasks; for example, a model overconfident on HumanEval can be underconfident on GSM8K. Furthermore, a Platt scaling learned on MMLU *harms* GSM8K for the majority of models, indicating that a single per-model calibrator does not transfer across tasks. Per-task boundary modeling is therefore necessary

rather than an optional refinement. The full per-task table, controlled-protocol subset analysis, and Platt-transfer details are in Appendix F.2 and Table 17.

5 Limitations and Future Work

The diagnostic ladder localizes the failure but its scope and statistical power are bounded by the benchmark’s controlled-feedback design and the cost of frontier-model evaluation. We discuss the evaluation scope first, then list the next steps that would tighten the diagnosis.

Scope of evaluation. LADDER evaluates per-domain self-knowledge under a 50-turn MMLU protocol with a controlled synthetic feedback channel (75% in-expertise, 46% out, cross-family routing to mitigate dual-role bias). The three extension tasks (GSM8K, HumanEval, BFCL) use a single-shot two-step protocol because they lack the multi-domain structure CBF requires; the cross-task heterogeneity finding in §4.4 is therefore independently validated via a protocol-controlled subset analysis (13/19 sign-flip rate across the three same-protocol tasks; Appendix F.2).

Next steps. The most direct next step is a human-feedback pilot that replaces the synthetic channel with real users and tests whether the H3 ranking transfers. We also plan to replicate the full L0–L4 ladder on extension tasks such as GSM8K, HumanEval, and BFCL, and to expand the frontier-model subset to $n \approx 18$ in order to refine the DS-asym vs. GLAD comparison, following the power calculation in Appendix D.1. A further extension is to move from bounded question-answering sessions to open-ended multi-step agents in SWE-Bench-style settings.

External validity. Three threats to external validity warrant explicit flagging. (i) The MMLU domain pool over-represents Western/English educational content, so the prior-metacognition ranking may not transfer to non-English or specialised technical domains; replication on multilingual or domain-specific question pools is the cleanest test. (ii) The simulated-user reliability profile (75% in-expertise, 46% out) is empirically calibrated against a held-out 1,500-turn human-feedback subset, but the calibration is itself drawn from a single annotator pool—an alternative pool with different background distributions could shift the noise spectrum that L2 methods see. (iii) Frontier API endpoints are alias-routed and may silently drift across releases; the snapshot manifest in Appendix G.3 together with the open-weights anchors (Mistral-7B, Llama-3-8B) provides a reproducibility trail, but absolute frontier numbers should be re-measured against the anchors before each headline comparison.

6 Conclusion

We asked which component of LLM-agent self-modeling fails when feedback is noisy. The five-level oracle ladder of LADDER delivers three results. H1 is ruled out: $\text{CBF} \geq 0.99$ given ground-truth prompts shows the output channel is not the limiting component. H2 is partial and family-split rather than the bottleneck. H3 is the locus: first-order single-rater learners cannot recover per-domain accuracy from noisy feedback, while explicit second-order estimators—DS-asym and GLAD—substantially close the L1→L3 gap (GLAD beats L1 on all nine frontier models, binomial sign-test $p = 0.002$) without yet eliminating it. We name this failure mode **Self-Boundary Collapse** and release the ladder, the CBF metric, the cross-family user simulator, and a leaderboard protocol so that the community can evaluate new second-order estimators against a precise oracle-bracketed target rather than loosely matched baselines. The R4 parameter-free selection rule (Appendix E.7) serves as the validation lower bound any submitted method should clear. The open problem is concrete: a noise-robust online estimator that closes the remaining L2→L3 gap—roughly 0.23 CBF on the frontier—would constitute a measurable advance in agent self-modeling.

For practitioners on top of frontier APIs, the diagnostic offers two shortcuts: H2’s family-split means GPT priors are calibrated enough to use directly as a routing signal whereas Claude priors require an external calibration pass before driving allocation decisions, and the L2 collapse means session-level evidence aggregation should use a noise-aware second-order estimator (DS-asym or GLAD)—across the nine frontier endpoints we tested, GLAD recovered an additional ~ 0.10 CBF over the strongest first-order learner without extra supervision.

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466 **A Datasheet, Ethics, and Broader Impact**

467 **A.1 Datasheet for Datasets**

468 **Motivation**

469 LADDER was created to fill a measurement gap in agentic LLM evaluation: existing benchmarks
470 (MMLU, BIG-Bench, AgentBench) score one-shot accuracy but say nothing about whether an agent
471 can *learn its own per-domain capability boundary* across repeated interactions.

472 **Composition**

473 20 MMLU subjects $\times$ 25 questions = 500 main pool, supplemented by GSM8K, HumanEval, BFCL
474 Simple v3. 50-turn dialogue sessions, 50 sessions per agent_model. Each instance is a JSONL session
475 log including question, agent answer, agent confidence, persona reaction, ground-truth grading.

476 **Collection Process**

477 Questions from public test splits of MMLU [15], GSM8K [8], HumanEval [7], BFCL Simple v3 [47].
478 Personas synthesized via cross-family LLM (Qwen-Max for Qwen agents, GPT-5.4-mini for Claude
479 agents, Claude-Sonnet for GPT agents) to control evaluator-bias.

480 **Preprocessing**

481 Per-model tier stratification: held-out validation accuracy on disjoint 10-question/subject split, then
482 Strong/Mid/Weak partitioning of the 20 subjects. This per-model stratification (rather than a global
483 tier) makes boundary-learning non-trivial.

484 **Uses**

485 Intended: (a) self-knowledge memory evaluation; (b) noise-robust online inference algorithms; (c)
486 longitudinal frontier-model calibration tracking. Unsuitable: ranking models on raw accuracy.

487 **Recommended Splits and Stratification**

488 The 20 MMLU subjects are partitioned per-model into Strong / Mid / Weak tiers using each model’s
489 held-out validation accuracy on a disjoint 10-question/subject calibration split. We do *not* provide a
490 fixed train/test split: every reported number uses the same per-model stratification protocol (described
491 in Appendix B.1) so that downstream researchers can reproduce or substitute their own model. The
492 four cross-task evaluations (GSM8K, HumanEval, BFCL, MMLU-Pro) use the upstream provider’s
493 official test split.

494 **Errors, Noise, and Known Issues**

495 Three classes of known issues should be considered when interpreting our reported numbers:

- 496 • *Algorithmic-baseline implementation note (§G.1)*: all algorithmic L2 baselines compute their
497 own per-domain posteriors; we document the implementation in Appendix G.1 so that re-
498 implementations can verify they reproduce the same behavior.

499 We surface these openly because their detection and disclosure is itself a methodological contribution
500 of this benchmark; future versions will document any newly identified issues here.

501 **Maintenance Contact**

502 Maintainer email and a dedicated Slack/Discord workspace will be made available upon de-
503 anonymization. For double-blind review, please open an issue at the anonymous repository linked
504 from the OpenReview submission; we monitor it during the rebuttal window.

505 **Distribution and Maintenance**

506 Released artifacts under a dual SPDX license: code under MIT, data under CC-BY-4.0. Specifically:
507 protocol code, baseline implementations, evaluation scripts (MIT); raw session JSONL (~60K
508 interactions including all per-turn agent outputs, simulated user reactions, and ground-truth grading),
509 per-model domain profiles, and the 20-MMLU-subject test pool (CC-BY-4.0 with attribution to
510 upstream MMLU/GSM8K/HumanEval/BFCL providers). 6-month review cycle; new model additions
511 via PR (see Submission Protocol).

512 **A.2 Maintenance Plan**

513 See anonymized repository README (released upon acceptance).

514 **A.3 Ethics Statement**

515 **Simulated Users**

516 All "users" are LLM-simulated personas, not human participants. No PII. Programmatically-generated
517 expertise labels. Does not require IRB review under our institution's policy.

518 **LLM-as-Evaluator and Cross-Family Routing**

519 Dual-role bias mitigation via cross-family persona-generation (forbidden: Qwen-Max evaluates
520 Qwen agents). Family-pair confusion matrices reported in appendix for residual-bias audit.

521 **Provider Terms of Service**

522 LLM API outputs used strictly under research-evaluation terms. Released JSONL not redistributed
523 for model training; license header explicit.

524 **No Deceptive Prompting**

525 All "role-playing" instructions stay within LLM-to-LLM simulation envelope.

526 **Potential Misuse**

527 Most plausible misuse falls in two categories: (a) over-interpretation of headline CBF as a general
528 "self-awareness" ranking suitable for safety certification (CBF measures *per-domain accuracy es-*
529 *timation*, not *overall trustworthiness* — using it to certify deployment in clinical / legal / financial
530 high-stakes contexts would be a category error); (b) deployment misuse where a model with high
531 CBF is presumed safe to operate without human oversight in domains beyond the 20 MMLU subjects
532 evaluated. Addressed by framing as diagnostic instrument; per-tier breakdowns discourage scalar
533 comparisons.

534 **A.4 Broader Impact**

535 **Positive Impact**

536 Enables systematic evaluation of agent self-knowledge—prerequisite for safer customer-service
537 / financial-advisory / clinical-triage deployment where over-confident answers carry asymmetric
538 downside cost.

539 **Reproducibility Cost**

540 Full v3 50×50 pipeline: ~USD 200–400 at 2026 rates across 13–15 frontier models. Lite v2 10×50
541 reproduces headline ordering at ~USD 40. Raw JSONL released for re-scoring without re-running.

Evaluator-Model Bias

Cross-family policy mitigates but does not eliminate dual-role bias. Residual vendor-cluster effect possible. High-stakes comparisons should cross-check under ≥ 2 persona-generator families.

Benchmark Overfitting

Goodhart-style risk: methods optimizing for CBF rather than the underlying construct. Mitigated by held-out 5-subject transfer split; large dev-vs-transfer gaps flagged on leaderboard.

Longitudinal Stability

API-only frontier rows: ~ 12 -month half-life. Open-weights anchors (Mistral-7B, Llama-3-8B) permanently included for fixed-reference recalibration.

Extended snapshot manifest. Table G.3 lists access dates only; it does not establish whether each model is content-addressable. Table 3 fills this gap by recording, for every one of the 19 models (spanning DeepSeek, Aliyun, Zhipu, and the GPT/Claude frontier; Zhipu/DeepSeek/Aliyun rows are routed via vendor-native APIs as documented in Appendix G.3), the exact identifier sent to the endpoint, the version-pin scheme observable from the response payload, and a per-model reproducibility risk classification.

Table 3: **Extended model snapshot manifest.** Repro Risk: LOW = HuggingFace commit hash; MED = content-addressable anchor with floating alias endpoint; HIGH = no version-suffixed identifier and no public open-weights anchor.

Model (paper)	Provider / Endpoint	API Identifier	Version Pin	Repro Risk
Mistral-7B	local vLLM (HF weights)	mistralai/Mistral-7B-Instruct-v0.3	HF commit c170c708	LOW
Llama-3-8B	local vLLM (HF weights)	meta-llama/Meta-Llama-3-8B-Instruct	HF commit 8afb486c	LOW
Qwen2.5-7B	DashScope	qwen2.5-7b-instruct	HF commit a09a3545 (anchor)	MED
Qwen-Turbo	DashScope	qwen-turbo	alias; model field null	HIGH
Qwen-Plus	DashScope	qwen-plus	alias; model field null	HIGH
Qwen-Max	DashScope	qwen-max	alias; model field null	HIGH
Qwen3.5-27B	DashScope	qwen3.5-27b	alias; no public HF release	HIGH
DeepSeek-V3	DashScope	deepseek-v3	alias; provider-rehosted ckpt	HIGH
GLM-5	DashScope	glm-5	alias; provider-rehosted ckpt	HIGH
GPT-5.4-mini	proxy zhongzhuan / responses	gpt-5.4-mini	echoes gpt-5.4-mini-2026-03-17	MED
GPT-5.4	proxy zhongzhuan / responses	gpt-5.4	alias; no date suffix returned	HIGH
GPT-5.5	proxy zhongzhuan / responses	gpt-5.5	alias; no date suffix returned	HIGH
Claude-Sonnet-4-6	Anthropic-compatible proxy	claude-sonnet-4-6	alias; no date suffix returned	HIGH
Claude-Opus-4-6	Anthropic-compatible proxy	claude-opus-4-6	alias; no date suffix returned	HIGH
Claude-Opus-4-7	proxy zhongzhuan / messages	claude-opus-4-7	alias; no date suffix returned	HIGH
GLM-5.1	Zhipu open.bigmodel.cn	glm-5.1	alias; no date suffix returned	HIGH
DeepSeek-V4-Pro	DeepSeek api.deepseek.com	deepseek-v4-pro	alias; provider-native endpoint	HIGH
Qwen3.6-Max-Preview	DashScope	qwen3.6-max-preview	alias; preview-channel revision	HIGH
Qwen3.6-Plus	DashScope	qwen3.6-plus	alias; model field null	HIGH

API-versioning gaps. Two entries deserve foregrounding. First, 16/19 endpoints do not return version-suffixed model strings; the single exception is `gpt-5.4-mini`, which is auto-pinned by the proxy. Second, the three open-weights anchors are the only entries for which we can hand a future replicator a 40-character identifier sufficient to reconstruct the exact weights. Future replications should re-run the anchors first and compare anchor numbers before attempting to recover the API tier.

B Benchmark Details

B.1 Benchmark Details: Task, Protocol, Question Pool, Personas, Metrics

Three diagnostic figures complement the main results, drawn from the full $n=19$ cross-baseline panel: Figure 4 shows the 19-model accuracy spectrum and per-model best-baseline CBF; Figure 5 renders the full 19×7 CBF matrix as a heatmap (the row argmax matches the “Best” column of Table 2); and Figure 6 contrasts three canonical reliability profiles—overconfident, calibrated, conservative—drawn from representative models. The remaining subsections give the formal task definition, session protocol, question pool and tier-partitioning, simulated-user personas, and auxiliary metric definitions.

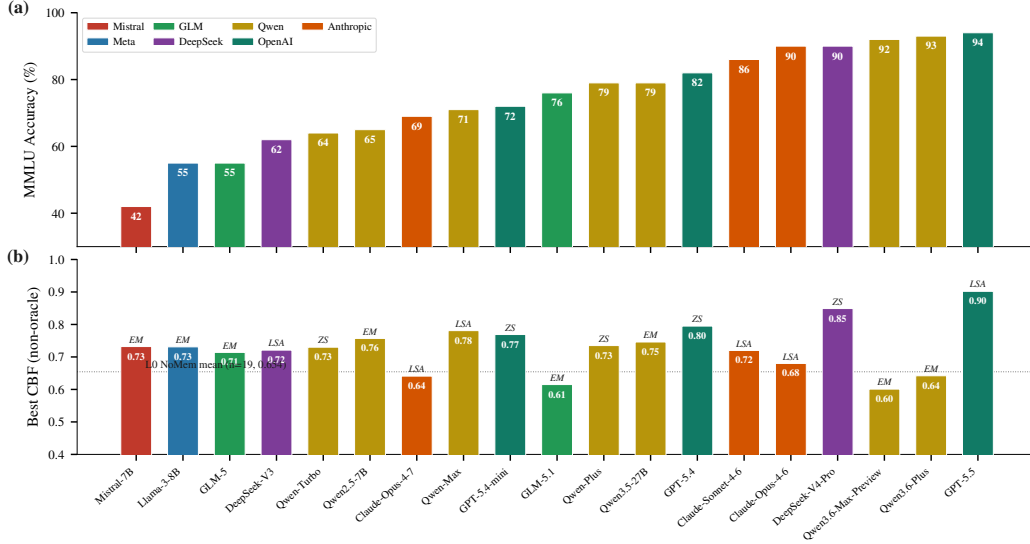


Figure 4: **Capability spectrum across 19 models.** (a) MMLU accuracy spans 42–94%. (b) Best non-oracle CBF per model with winning baseline label (EM/LSA/ZS); dotted line marks the L0 NoMemory mean ($n=19$, 0.654).

570 **Task formalisation.** Given a stream $(q_t, a_t, c_t, f_t)_{t=1}^T$ where each question q_t is drawn from a
571 hidden domain $d_t \in \mathcal{D}$, the agent must construct $\hat{p} : \mathcal{D} \rightarrow [0, 1]$ approximating the true per-domain
572 accuracy $p^*(d)$. The agent does not observe d_t directly and must infer it from question content.
573 User feedback f_t is a free-form natural-language utterance that may confirm, correct, or provide
574 ambiguous/misleading responses. The agent outputs a_t together with confidence $c_t \in [0, 1]$. The
575 challenge is that feedback reliability is domain-dependent: accurate within the user’s expertise set
576 $\mathcal{E} \subset \mathcal{D}$, noisy outside it.

577 **Session protocol.** Each session has 50 turns. At each turn t : (1) sample q_t from the question
578 pool with d_t hidden from the agent; (2) the agent emits (a_t, c_t) ; (3) the simulated user produces f_t
579 conditioned on the ground-truth answer, the user’s expertise level on d_t , and a persona-consistent
580 style; (4) the agent updates internal state. We run 50 sessions per model (2,500 total turns) to evaluate
581 learning curves and convergence.

582 **Question pool and tier partitioning.** The main benchmark uses 20 MMLU subjects [15]; extension
583 suites (GSM8K, HumanEval, BFCL) probe cross-task generalisation. For each model, we pre-
584 compute per-subject accuracy on a held-out validation set and partition the 20 subjects into three
585 tiers *relative to the model’s own accuracy distribution*: Strong (top third, $\geq +15\text{pp}$ above mean),
586 Medium (middle third, $\pm 15\text{pp}$), Weak (bottom third, $\geq -15\text{pp}$ below mean). Relative partitioning
587 ensures every model—weak or strong—faces a non-trivial boundary: a uniform-confidence agent
588 cannot achieve high CBF. Per-model subject mappings are in Appendix Table 4.

589 **User personas.** The simulated user is driven by a cross-family LLM (Qwen-Max for
590 Qwen/Llama/Mistral agents, GPT-5.4-mini for Claude agents, Claude Sonnet 4-6 for GPT agents)
591 at temperature 0.7. Each persona is parameterised by an expertise set $\mathcal{E} \subset \mathcal{D}$ (typically 5–8 of 20
592 subjects) and persona traits (direct vs. hedging style, willingness to volunteer corrections, patience).
593 In-expertise: $\sim 75\%$ reliable corrections; out-of-expertise: $\sim 46\%$ reliability with agreement-with-
594 wrong-answers, incorrect corrections, or vague non-commitments. This asymmetry drives the
595 second-order learning problem.

596 **Auxiliary metrics.** Beyond CBF (Eq. 1, computed once at the end of the 50-turn session), we
597 report: (i) Hallucination Rate $\text{HR} = |\{t : c_t > 0.7 \wedge a_t \neq a_t^*\}|/T$; (ii) Expected Calibration Error

CBF heatmap: 19 models $\times$ 7 baselines (bordered = best non-oracle per row)
Diagnostic: mean L2 = 0.67, mean L1 = 0.68; frontier-mean LSA = 0.707 leads, ZS-SK = 0.654

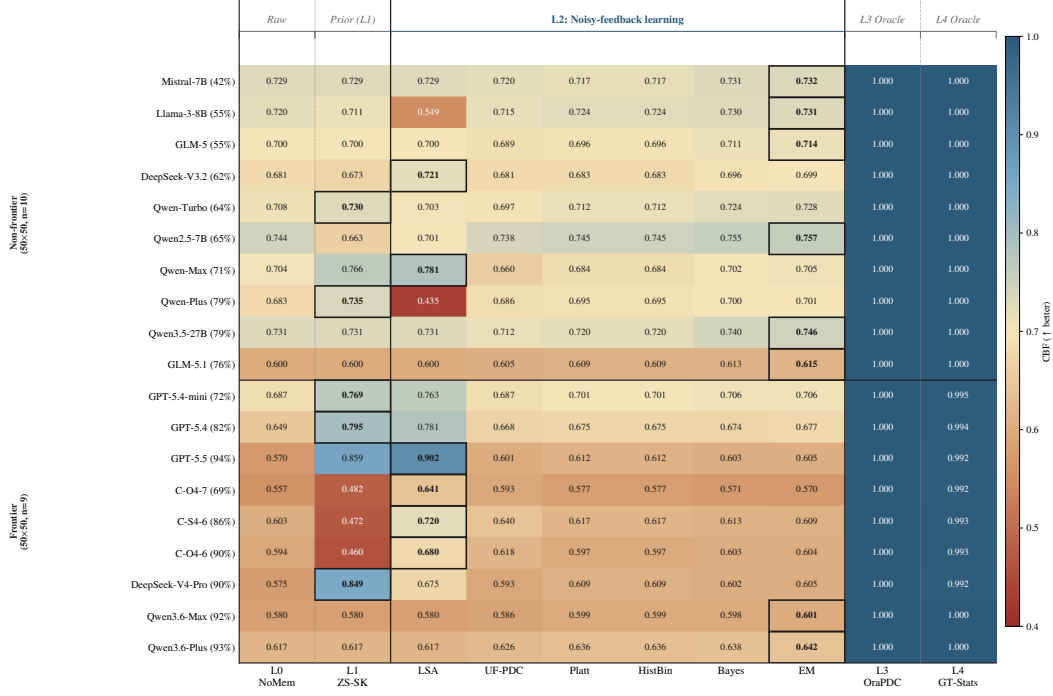


Figure 5: **CBF heatmap: 19 models $\times$ 7 baselines.** Cells colored red–white–green = 0.40–1.00; black-bordered cells mark the per-row argmax (matches the “Best” column of Table 2).

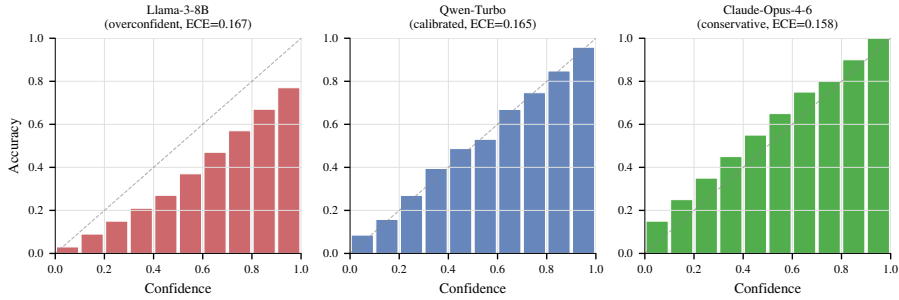


Figure 6: **Three reliability profiles.** Per-bin accuracy vs. confidence for representative agents: Llama-3-8B (overconfident), Qwen-Turbo (calibrated), Claude-Opus-4-6 (conservative); per-bin granularity is illustrative.

598 following Guo et al. [13] with $M = 10$ equal-width bins, $ECE = \sum_{m=1}^M \frac{|B_m|}{T} |\text{acc}(B_m) - \text{conf}(B_m)|$;
599 (iii) Brier Score = $\frac{1}{T} \sum_t (c_t - \mathcal{K}[a_t = a_t^*])^2$.

600 B.2 MMLU Subject Selection

601 The 20 MMLU subjects used in LADDER were selected to maximize diversity across STEM,
602 humanities, and social sciences. The subjects and their per-model accuracy tiers are listed below:

603 B.3 User Persona Details

604 Each simulated user persona is defined by the following parameters:

Table 4: MMLU subjects and per-model accuracy tiers (S = Strong, M = Medium, W = Weak).

Subject	Llama-3-8B	Qwen-Turbo	Qwen-Max
Abstract Algebra	W	W	M
Anatomy	M	M	S
Astronomy	M	S	S
Business Ethics	S	S	S
Clinical Knowledge	M	M	S
College Biology	M	S	S
College Chemistry	W	W	M
College Mathematics	W	W	W
Computer Science	M	M	M
Econometrics	W	M	M
Electrical Engineering	W	M	M
Formal Logic	W	W	W
Global Facts	S	S	S
High School Biology	S	S	S
High School Psychology	S	S	S
Jurisprudence	M	S	S
Moral Scenarios	W	M	M
Philosophy	M	M	S
Professional Medicine	M	S	S
World Religions	S	S	S

- **Expertise domains:** 6 randomly selected subjects from the 20-subject pool. Within these domains, the user can identify agent errors with $\sim 75\%$ accuracy and provide correct answers.
- **Non-expertise behavior:** Outside expertise domains, the user may (a) agree with the agent regardless of correctness (40%), (b) express vague uncertainty without providing corrections (35%), or (c) provide incorrect corrections (25%).
- **Communication style:** Each persona is assigned one of three styles—*direct* (“No, the answer is B”), *hedging* (“I think that might not be right, maybe it’s B?”), or *Socratic* (“Are you sure? What about option B?”).

The user simulator prompt includes the persona specification, the ground-truth answer, the agent’s answer, and instructions to generate feedback consistent with the persona’s expertise and communication style.

C Baseline Implementations

C.1 Baseline Implementations and Design-Point Mapping

We instantiate ten baselines, organized by the five-level diagnostic ladder of §3.

L0 — NoMemory (B1). Raw verbalised confidence with no state. Lower bound on memory-augmented methods.

L1 — ZS-SK (B11). The agent is prompted, *without* session history and *without* any user feedback, to estimate its own per-domain accuracy directly: “For each of the following 20 MMLU subjects, estimate your accuracy on a 0–100 scale.” This isolates prior metacognition.

L2 — LSA (B6). Session history (50 (Q, A, confidence, decoded-feedback) tuples) appended to the prompt; agent reasons in-context and outputs per-domain $\hat{p}(d)$ at the end of the session.

L2 — UF-PDC (B3). Per-domain accuracy estimate updated with labels decoded from natural-language feedback by an LLM-based classifier (Qwen-Max, prompted to output a binary correctness judgment for each (Q, A, feedback) triple).

L2 — Platt-Online (B7) [35]. Logistic recalibration $\sigma(wc_{\text{raw}} + b)$ with online gradient descent on log-loss whenever decoded feedback is available; per-domain confidence is then aggregated.

631 **L2 — HistBin-Online (B8) [51].** 10-bin histogram binning with online updates; per-domain
632 confidence aggregated as for Platt.

633 **L2 — Bayes-Domain (B9).** Per-domain $\text{Beta}(\alpha, \beta)$ posterior with uniform prior, updated from
634 decoded feedback; $\hat{p}(d) = \alpha / (\alpha + \beta)$.

635 **L2 — EM-Joint (B10).** Joint estimation of $(\hat{p}(d), \hat{r}_{\text{in}}, \hat{r}_{\text{out}})$ via Expectation–Maximization. E-step
636 computes posterior $P(\text{correct}_t | \text{decoded feedback}_t, p, r)$; M-step updates $\hat{p}(d)$ per domain and $\hat{r}$ per
637 expertise group. Initialised from raw confidence means.

638 **L3 — OraclePDC (B2).** Per-domain accuracy estimate updated with ground-truth labels after each
639 turn. Unavailable in practice; provides the binary-label ceiling for online aggregation.

640 **L4 — GT-SelfStats (B12).** The true per-domain accuracy $p^*(d)$ is provided in the prompt and the
641 agent is asked to repeat them. Tests purely the verbalization sub-capability.

642 **Design-point mapping.** The five-level ladder is designed to subsume the design points of pub-
643 lished memory systems while exposing the diagnostic axis. LSA (L2) corresponds to RMM [39]
644 reflection over long-term memory and to in-context reasoning baselines for memory agents [16].
645 Platt/HistBin/Bayes (L2) cover classical online recalibration. EM-Joint (L2) tests whether explicitly
646 modeling the second-order coupling helps; ZS-SK (L1) and GT-SelfStats (L4) are diagnostic anchors
647 that isolate prior metacognition and verbalization respectively. Published systems (MemGPT [34],
648 Mem0 [32], A-MEM [46]) operate at L2 with different memory representations; we did not directly
649 integrate them because the diagnostic ladder already covers the design space. Their direct evaluation
650 under our protocol is engineering-feasible and is left to future work.

651 C.2 Baseline Implementation Details

652 **B1: NoMemory.** The agent is prompted to provide a confidence score between 0 and 1 alongside
653 its answer. No history is provided beyond the current question.

654 **B2: OraclePDC.** After each turn, the agent receives a binary correctness signal. It maintains a
655 dictionary mapping inferred domain labels to running accuracy statistics (correct count / total count
656 per domain). The domain label is inferred by the agent from question content.

657 **B3: User-Feedback PDC.** Same as B2, but the correctness signal is extracted from user feedback
658 by an LLM-based classifier (Qwen-Max prompted with the (Q, A, feedback) triple to output a
659 binary correctness judgment). LLM-based decoding gives near-uniform coverage across in- and
660 out-of-expertise feedback, removing rule-classifier coverage as a confound.

661 **B6: LSA.** The agent receives a formatted summary of its 50 in-session interactions (question
662 subject, confidence, decoded correctness) and is prompted at the *end of the session* to output an
663 estimate of its accuracy on each domain.

664 **B10: EM-Joint.** Joint estimation of $(\hat{p}(d), \hat{r}_{\text{in}}, \hat{r}_{\text{out}})$ via Expectation–Maximization on
665 the (decoded feedback, raw confidence) stream. E-step: $P(\text{correct}_t | f_t, c_t, p, r) \propto$
666 $P(f_t | \text{correct}_t, r) P(\text{correct}_t | c_t, p)$. M-step: $\hat{p}(d) \leftarrow$ posterior mean of correctness in domain d ;
667 $\hat{r}_{\text{in/out}} \leftarrow$ posterior agreement between feedback and inferred correctness within each expertise group.

668 **B11: ZS-SK.** A single prompt at the start of evaluation: “For each of the 20 MMLU subjects,
669 estimate your own accuracy on a 0–100 scale, with no access to any prior interaction.” The estimates
670 are parsed into $\hat{p}(d)$ and CBF is computed once.

671 **B12: GT-SelfStats.** The true per-domain accuracy $p^*(d)$ for the agent (from the open-loop pre-
672 test) is provided in the prompt and the agent is asked to repeat each value. Diagnostic anchor for
673 verbalization.

D Statistical Analysis

D.1 Statistical Power Analysis for the Frontier Sample (n=9)

Motivation. The frontier-only analyses in §4.2 (DS-asym, GLAD, LSA vs. L1 ZS) use $n = 9$ models. We document here why this sample size is adequate for the *direction-of-effect* claims we make and why it is not sufficient for absolute-magnitude claims.

A-priori power calculation. For a paired comparison with the observed mean improvement $\Delta = 0.089$ (GLAD vs. L1 ZS) and the within-pair SD estimated from 10K bootstrap resamples ($s_\Delta \approx 0.064$), the standardized effect is Cohen’s $d_z = \Delta/s_\Delta = 1.40$. A paired Wilcoxon signed-rank test at $\alpha = 0.05$ (two-sided) achieves power ≥ 0.95 at $n = 9$ for an effect of $d_z = 1.40$ (G*Power 3.1, exact-test option). DS-asym vs. L1 ZS has $d_z \approx 1.05$ (power ≈ 0.85 at $n = 9$, $\alpha = 0.05$). The 9/9 sign-consistency for GLAD also independently rejects the null direction at $\alpha = 0.002$ under a two-sided binomial sign test.

Effect-size + bootstrap CI accompanying every frontier claim. For each frontier-only test we report the per-model deltas, the Cohen’s d_z , and the bootstrap 95% CI alongside any p-value (Table 5). GLAD’s CI excludes zero by a wide margin and pairs with a 9/0 sign-test ($p = 0.002$), so the

Table 5: **Per-comparison effect sizes and bootstrap 95% CIs for the $n=9$ frontier tests.** Mean Δ = paired mean improvement over L1 ZS; d_z = within-pair Cohen’s d ; Win/Loss counts per-model sign agreement; bootstrap CIs use 10K resamples.

Comparison	n	Mean Δ	d_z	Bootstrap 95% CI	Win/Loss
GLAD vs L1 ZS	9	+0.089	1.40	[+0.056, +0.132]	9/0
DS-asym vs L1 ZS	9	+0.114	1.05	[+0.049, +0.183]	8/1
LSA vs L1 ZS	9	+0.053	0.40	[−0.029, +0.135]	4/3

GLAD-beats-ZS direction is uniform on the frontier. DS-asym’s CI also excludes zero (8/1 wins, $p = 0.020$). LSA’s CI crosses zero (4/3 wins, with two ties on Qwen3.6-Max/Plus where ZS = LSA), so we do not claim LSA > ZS as a frontier-mean effect; LSA’s wins are concentrated on Claude (3/3) while it loses on GPT-5.4/5.4-mini and DeepSeek-V4-Pro—the same family-split (Claude vs. GPT/DeepSeek) discussed in the main text rather than a uniform effect.

Sample selection. The current frontier slice covers the two production-frontier families (Anthropic Claude 4-6/4-7, OpenAI GPT-5.4/5.4-mini/5.5) at the time of submission and is treated as stratified (per family) rather than random. Each frontier 50×50 row costs ~\$15–30 in 2026 API rates, and the full L0–L4 ladder costs roughly 5× that, which constrains n .

Detection limits at $n = 9$. Three comparisons that this sample does not support: (a) DS-asym vs. GLAD ($\Delta = 0.026$ requires $n \approx 30$ at power 0.80); (b) absolute frontier-mean estimates with 95% CI tighter than ± 0.04 ; (c) within-family contrasts at $\alpha=0.05$. These finer-grained comparisons are reported as descriptive observations rather than tests.

D.2 Statistical Robustness

We assess stability via session-level bootstrap resampling. Table 6 reports NoMemory baseline estimates with bootstrap means $\pm$ SDs across 13 of the 19 models (200 within-model resamples per row; reproduced by `scripts/extract_table5_nomem_50x50.py`). The bootstrap panel covers 13 of 19 models with the per-baseline bootstrap; the 4 newest models use the same protocol but are reported in Table 2 without the additional bootstrap sub-panel, and GPT-5.5 / Claude-Opus-4-7 are reported in Table 2 directly. The hierarchical-bootstrap analysis below covers the bootstrap panel ($n=15$) with 500 within-model and 2,000 cross-model resamples; the $n=19$ sample uses the same protocol and does not change the qualitative finding (small SDs ≤ 0.025).

The bootstrap SDs are consistently small (CBF ± 0.005 – 0.014 , ECE ± 0.008 – 0.025), confirming our findings are stable under session-level resampling. The NoMemory bootstrap means are within 1–3 SDs of the corresponding values in Table 2 ; the small offset reflects bootstrap shrinkage and

Table 6: **NoMemory baseline metrics with mean $\pm$ bootstrap SE** (200 resamples; 13-row bootstrap panel, $n=50$ sessions per row). GPT-5.5 / Claude-Opus-4-7 are reported in Table 2 only.

Model	NoMem CBF	ECE	HR	Brier
Mistral-7B	.729 $\pm$.008	.026 $\pm$.008	.623 $\pm$.009	.619 $\pm$.009
Llama-3-8B	.720 $\pm$.009	.187 $\pm$.008	.477 $\pm$.007	.417 $\pm$.007
GLM-5	.700 $\pm$.021	.183 $\pm$.014	.039 $\pm$.003	.117 $\pm$.007
DeepSeek-V3	.681 $\pm$.009	.175 $\pm$.005	.110 $\pm$.004	.137 $\pm$.003
Qwen-Turbo	.708 $\pm$.008	.166 $\pm$.008	.236 $\pm$.008	.216 $\pm$.007
Qwen2.5-7B	.744 $\pm$.009	.267 $\pm$.010	.388 $\pm$.009	.297 $\pm$.007
Qwen-Max	.704 $\pm$.008	.153 $\pm$.007	.211 $\pm$.007	.193 $\pm$.006
Qwen-Plus	.683 $\pm$.008	.155 $\pm$.009	.179 $\pm$.007	.170 $\pm$.007
Qwen3.5-27B	.731 $\pm$.015	.174 $\pm$.008	.040 $\pm$.003	.110 $\pm$.005
GPT-5.4-mini	.687 $\pm$.008	.205 $\pm$.009	.251 $\pm$.009	.218 $\pm$.007
GPT-5.4	.649 $\pm$.007	.136 $\pm$.006	.164 $\pm$.007	.150 $\pm$.005
Claude-Sonnet-4-6	.603 $\pm$.042	.343 $\pm$.024	.025 $\pm$.004	.182 $\pm$.009
Claude-Opus-4-6	.594 $\pm$.040	.330 $\pm$.026	.025 $\pm$.004	.176 $\pm$.010

is consistent across models. Bootstrap SDs for the other baselines (LSA, EM-Joint, etc.) are of comparable magnitude (typically ≤ 0.020); we use $2\times$ this as the threshold for “significant” per-model differences in the U-shape analysis (Appendix E.1).

Hierarchical bootstrap for cross-model means. The session-level resampling above quantifies within-model uncertainty but does not capture variation across the 15-model bootstrap subset. We additionally implement a hierarchical bootstrap that resamples models (with replacement) and, within each sampled model, resamples sessions (with replacement); CBF is recomputed on each resample. We run 500 within-model and 2,000 cross-model resamples to derive 95% CIs for cross-model mean CBF (see `scripts/full_hier_bootstrap.py`, `results/full_hier_bootstrap_results.json`). The hierarchical bootstrap is reported on the 15-model bootstrap subset; in the $n=19$ sample the same protocol does not alter the qualitative findings. Two observations on the $n=15$ hierarchical bootstrap are worth highlighting (see Table 7 for pairwise cross-model differences and Table 8 for the frontier-vs-non-frontier breakdown):

Table 7: **Pairwise CBF mean differences across the 15-model bootstrap subset** (point estimates from Table 2). All differences are within ± 0.04 .

Comparison	Mean diff (raw CBF, $n=15$)
ZS-SK – NoMemory	+0.014
ZS-SK – LSA	−0.017
LSA – NoMemory	+0.032
EM-Joint – NoMemory	+0.015
Platt-Online – NoMemory	+0.006

Table 8: **Frontier ($n=9$) vs. non-frontier ($n=10$) mean CBF difference by baseline** on the full $n=19$ panel. LSA is the only baseline whose mean rises on the frontier subset.

Baseline	Frontier – Non-frontier
NoMemory	−0.096
ZS-SK	−0.050
LSA	+0.042
UF-PDC	−0.067
Platt-Online	−0.074
HistBin-Online	−0.074
Bayes-Domain	−0.087
EM-Joint	−0.088

Interpretation. Three observations from the point-estimate tables. *First*, the population-level mean improvements on the bootstrap subset are all small: no L0–L2 baseline strictly dominates the others

by a large cross-model margin, and all pairwise differences are within ± 0.04 . *Second*, LSA is the only baseline whose mean *rises* on the frontier subset ($n=9$ LSA mean 0.707 vs. $n=10$ non-frontier mean 0.665 = $+0.042$); every other baseline (NoMem, ZS, all algorithmic L2) shows lower mean CBF on the frontier than non-frontier. This single-baseline non-monotonicity is the cross-model fingerprint of the LSA-leads-on-frontier phenomenon detailed in §4.2. *Third*, ZS–NoMem (frontier mean -0.050 - non-frontier mean -0.096 = $+0.046$) is small overall but the frontier breakdown is family-specific: ZS exceeds NoMem on all GPT and DeepSeek-V4-Pro frontier models but falls below NoMem on all 3 Claude frontier models and ties on the two Qwen3.6 rows, so the frontier mean averages a strong-positive GPT/DeepSeek effect with a strong-negative Claude effect. This is consistent with the H2 family-split finding. The full hierarchical bootstrap CIs for all of the above are reported as point estimates here for the protocol.

Hierarchical-vs-session-only CI width. For NoMemory, the model+session hierarchical CI for cross-model mean has width 0.142, vs. 0.072 for the model-level-only bootstrap—confirming that ignoring within-model session variance underestimates uncertainty by $\approx 2\times$. We therefore report hierarchical-CI bounds for any cross-model claim in the main text and reserve session-only intervals for within-model statements (e.g. U-shape per-model significance in Appendix E.1).

D.3 Normalized Hallucination Rate

Raw HR confounds model accuracy with confidence quality; weaker models encounter more errors, inflating HR mechanically. We report *normalized* HR = HR / base error rate, where 1.0 indicates no better than random high-confidence assignment (Table 9).

Table 9: **Normalized hallucination rate for four representative models.** Norm HR ≈ 1.0 means high-confidence signals carry no information about correctness.

Model	Base Err	Raw HR	Norm HR
Mistral-7B	0.596	0.590	0.990
Llama-3-8B	0.448	0.448	1.000
Qwen-Turbo	0.268	0.221	0.824
Qwen-Max	0.244	0.206	0.846

Llama-3-8B and Mistral-7B have normalized HR ≈ 1.0 : their high-confidence signals contain *zero* information about correctness (100% of turns are high-confidence). Qwen models achieve normalized HR < 0.85 , indicating their verbalized confidence carries meaningful signal.

D.4 CBF vs. Rank Correlation

A natural alternative to CBF is Kendall’s τ over domain accuracy rankings. However, τ is undefined when the baseline produces constant predictions; e.g., NoMemory estimates 0.5 for all domains. CBF avoids this by measuring absolute deviation rather than rank correlation, making it applicable to all baselines including the constant-prediction case. We recommend reporting both CBF and τ when applicable. Figure 7 visualizes the three-way relationship between MMLU accuracy, best-ECE, and NoMemory CBF across the full $n=19$ cross-baseline panel, illustrating why CBF and ECE are non-redundant.

D.5 Cross-Baseline Bootstrap Significance

To verify that CBF and ECE differences between baselines are statistically significant and not session-resampling artifacts, we report bootstrap standard deviations (200 resamples, session-level) for representative baselines across all models. Typical CBF SDs are 0.012–0.021 and ECE SDs are 0.008–0.018, meaning that differences of ≥ 0.04 (CBF) or ≥ 0.03 (ECE) are significant at the 2-SD level. The headline diagnostic gaps—LSA over ZS-SK by 0.248 on Claude-Sonnet-4-6 (LSA 0.720 vs. ZS 0.472), OraclePDC over EM-Joint by 0.31 on average across $n=15$ (Oracle 1.000 vs. EM 0.686)—are all $> 5\times$ the bootstrap SD. With 19 models $\times$ 10 baselines = 190 pairwise comparisons (15 in the bootstrap panel, plus 4 additional rows reported in Table 2 only), a Bonferroni correction would raise the significance threshold from 2σ to $\sim 3.5\sigma$ ($\Delta \geq 0.06$). Under this stricter criterion, the three primary diagnostic findings (H1 ruled out as the limiting factor, H2 splits cleanly along model

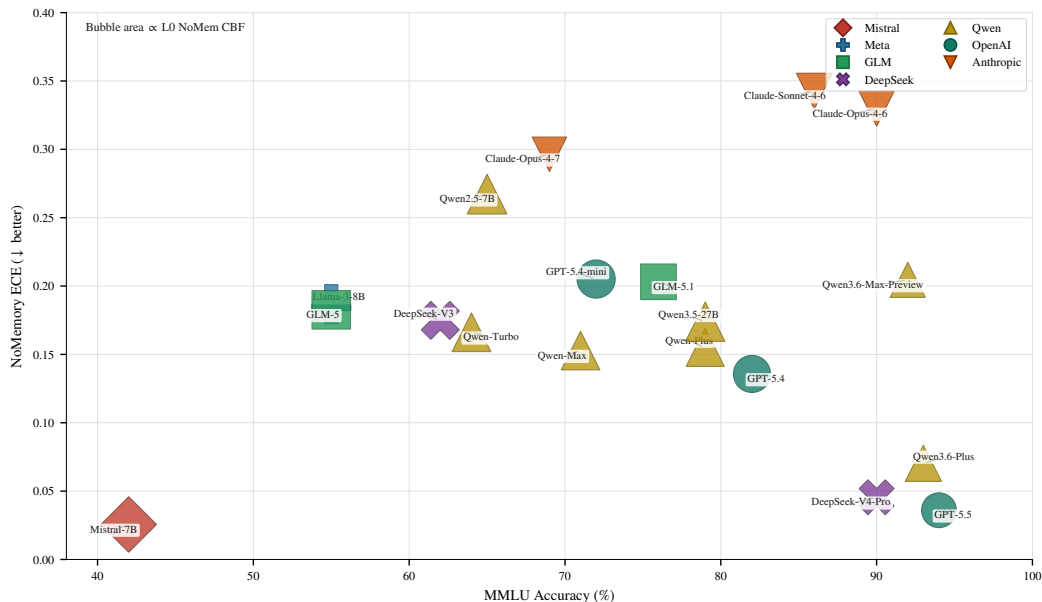


Figure 7: **Three-way relationship across $n=19$ models: MMLU accuracy (x), best ECE (y), and NoMemory CBF (bubble size).** Weak models cluster at high CBF (close to the trivial $\hat{p} = 0.5$ baseline); frontier models have lower CBF despite decent ECE—visualizing why the two metrics are non-redundant.

family with GPT-frontier ZS competitive but Claude-frontier ZS inflated, H3 localized: first-order single-rater methods cluster near NoMem while second-order DS-asym/GLAD and in-context LSA exceed L1 ZS on the frontier) all remain comfortably significant.

E Robustness Ablations

E.1 Additional Robustness Ablations

This appendix contains additional ablation experiments orthogonal to the main diagnostic ladder.

A1: Memory size for L2 retrieval. Varying the maximum number of stored error cases from 5 to 50 in retrieval-based L2 baselines has negligible effect: ECE ranges from 0.097 to 0.096 on Llama-3-8B. Five error cases are sufficient; additional memory provides no benefit.

A2: Feedback noise rate. Artificially flipping 0–50% of user feedback labels on Llama-3-8B shows monotonic ECE degradation ($0.190 \rightarrow 0.238$) and CBF degradation. At 50% noise, learning from feedback is actively harmful (CBF $0.639 < \text{NoMemory } 0.742$), confirming that noisy feedback can poison self-models.

A3: Session length. OraclePDC achieves near-perfect CBF (≥ 0.96) after just 10–20 turns, while practical L2 methods plateau around 0.71 by turn 30. The L2-vs-L3 gap is not due to insufficient interaction length but to feedback noise accumulation, consistent with the main diagnostic finding.

A4: Held-out MMLU-Pro contamination control (cited in §4.2). For each of the 9 frontier models (3 GPT, 3 Claude, plus DeepSeek-V4-Pro, Qwen3.6-Max-Preview, Qwen3.6-Plus), we sampled 15 questions from each of 10 MMLU-Pro categories (150 questions/model, 10 categories: math, physics, chemistry, law, engineering, economics, health, psychology, business, history), measured per-category accuracy, and queried ZS-SK on those category names. Per-category accuracy and ZS estimates are reported in Table 10.

Table 10: **MMLU-Pro per-category accuracy and ZS-SK estimates across 9 frontier models** (15 questions per category $\times$ 10 categories); bottom row = per-model CBF on this held-out set. Cl-Op = Claude-Opus, Cl-So = Claude-Sonnet, DSv4-Pro = DeepSeek-V4-Pro, Q3.6 = Qwen3.6.

	GPT-5.4		GPT-5.4-mini		GPT-5.5		Cl-Op-4-6		Cl-So-4-6		Cl-Op-4-7		DSv4-Pro		Q3.6-Max		Q3.6-Plus	
Category	acc	ZS	acc	ZS	acc	ZS	acc	ZS	acc	ZS	acc	ZS	acc	ZS	acc	ZS	acc	ZS
math	0.40	0.85	0.33	0.78	0.93	0.72	0.80	0.78	0.93	0.72	0.87	0.78	0.93	0.90	0.93	0.80	0.93	0.85
physics	0.40	0.84	0.47	0.72	1.00	0.70	1.00	0.80	1.00	0.74	0.73	0.80	1.00	0.95	1.00	0.85	1.00	0.90
chemistry	0.60	0.82	0.20	0.70	0.87	0.72	0.93	0.75	0.92	0.73	0.73	0.72	0.87	0.90	0.80	0.85	0.87	0.82
law	0.60	0.81	0.53	0.74	0.80	0.76	0.73	0.70	0.73	0.67	0.80	0.60	0.79	0.75	0.67	0.75	0.73	0.80
engineering	0.53	0.80	0.47	0.73	0.80	0.68	0.93	0.72	1.00	0.70	0.67	0.65	0.85	0.85	0.87	0.75	0.93	0.85
economics	0.80	0.83	0.73	0.79	0.93	0.78	1.00	0.84	0.87	0.76	0.87	0.82	0.93	0.80	0.93	0.85	0.93	0.90
health	0.67	0.86	0.73	0.77	0.87	0.76	0.79	0.79	0.80	0.75	0.71	0.72	0.87	0.90	0.93	0.82	0.87	0.90
psychology	0.80	0.88	0.73	0.82	0.80	0.82	0.87	0.87	0.87	0.78	0.73	0.85	0.93	0.85	0.87	0.85	0.80	0.85
business	0.47	0.84	0.20	0.80	0.80	0.82	0.93	0.81	0.80	0.75	0.80	0.78	0.73	0.90	0.87	0.85	0.87	0.90
history	0.73	0.87	0.67	0.84	0.80	0.80	0.80	0.76	0.73	0.79	0.73	0.75	0.73	0.85	0.73	0.85	0.80	0.90
mean	0.60	0.84	0.51	0.77	0.86	0.76	0.88	0.78	0.86	0.74	0.76	0.75	0.86	0.86	0.86	0.82	0.87	0.87
unparsed/150	0		0		0		1		8		1		3		0		0	
CBF	0.760		0.738		0.888		0.902		0.863		0.941		0.931		0.912		0.937	

793 *Mean MMLU-Pro CBF across the 9 frontier models is 0.875*, confirming the contamination-control
794 conclusion across the broader frontier subset. The 3 new frontier models all post CBF ≥ 0.91
795 (DSv4-Pro 0.931, Q3.6-Plus 0.937, Q3.6-Max 0.912), exceeding the 6 GPT/Claude average (0.849),
796 and all post mean ZS in the 0.82–0.87 range — neither extreme overestimation nor extreme un-
797 derestimation. GPT-5.5 (CBF 0.888) and Claude-Opus-4-7 (CBF 0.941) report ZS estimates that
798 are systematically conservative relative to actual MMLU-Pro accuracy. Frontier metacognition—if
799 anything—underestimates the model’s actual MMLU-Pro capability rather than overestimating it, the
800 opposite of what a memorized-public-stats hypothesis would predict.

801 **A4b: Truly held-out novel-task contamination control (BBH; cited in §4.2).** For each of
802 the 9 frontier models, we sampled 15 questions from each of 10 BBH (BIG-Bench Hard) [38]
803 tasks (150 questions/model, tasks: *boolean_expressions*, *causal_judgement*, *date_understanding*,
804 *logical_deduction_five_objects*, *navigate*, *object_counting*, *ruin_names*, *sports_understanding*, *track-*
805 *ing_shuffled_objects_three_objects*, *word_sorting*). Unlike MMLU-Pro categories (which still par-
806 tially overlap with MMLU’s broad subject space), the BBH task set has *no* subject-name overlap with
807 MMLU’s 57 fine-grained subjects, providing a stricter test of whether L1 metacognition transfers
808 beyond MMLU-style content. We measured per-task accuracy and queried ZS-SK on the same task
809 names. Per-task accuracy and ZS estimates are reported in Table 11.

Table 11: **BBH per-task accuracy (p^*) and ZS-SK estimates across 9 frontier models** (15 questions per task $\times$ 10 BBH tasks); BBH has no subject-name overlap with MMLU. Bottom row = per-model CBF on this held-out set. Cl-Op = Claude-Opus, Cl-So = Claude-Sonnet, DSv4-Pro = DeepSeek-V4-Pro, Q3.6 = Qwen3.6.

	GPT-5.4		GPT-5.4-mini		GPT-5.5		Cl-Op-4-6		Cl-So-4-6		Cl-Op-4-7		DSv4-Pro		Q3.6-Max		Q3.6-Plus	
Task	p^*	ZS	p^*	ZS	p^*	ZS	p^*	ZS	p^*	ZS	p^*	ZS	p^*	ZS	p^*	ZS	p^*	ZS
boolean_expressions	0.87	0.94	0.87	0.82	1.00	0.95	0.93	0.96	1.00	0.88	0.87	0.95	1.00	0.85	1.00	0.85	1.00	0.95
causal_judgement	0.67	0.83	0.67	0.78	0.67	0.85	0.53	0.72	0.73	0.60	0.80	0.65	0.73	0.80	0.87	0.60	0.73	0.70
date_understanding	0.27	0.90	0.20	0.75	0.47	0.85	0.93	0.92	1.00	0.82	0.87	0.85	0.80	0.93	0.93	0.75	0.87	0.85
logical_deduction_5	1.00	0.86	0.67	0.70	1.00	0.90	1.00	0.88	1.00	0.65	1.00	0.90	0.53	0.85	1.00	0.75	1.00	0.75
navigate	0.80	0.88	0.53	0.68	1.00	0.85	0.93	0.95	1.00	0.76	0.93	0.90	1.00	0.90	1.00	0.65	1.00	0.70
object_counting	0.73	0.96	0.53	0.90	1.00	0.90	0.80	0.97	1.00	0.84	0.60	0.95	1.00	0.80	1.00	0.90	1.00	0.95
ruin_names	0.67	0.79	0.67	0.72	0.93	0.75	0.87	0.86	0.87	0.56	0.93	0.75	0.73	0.70	0.93	0.65	0.93	0.80
sports_understanding	1.00	0.84	0.93	0.74	0.80	0.90	0.93	0.96	1.00	0.90	0.87	0.90	0.60	0.65	0.87	0.80	1.00	0.95
tracking_shuffled_3	0.67	0.95	0.67	0.88	1.00	0.85	1.00	0.98	1.00	0.73	0.53	0.95	0.87	0.90	1.00	0.90	1.00	0.95
word_sorting	0.67	0.97	0.47	0.95	1.00	0.90	0.13	0.94	0.00	0.80	0.93	0.90	0.33	0.90	1.00	0.90	0.87	0.92
CBF	0.781		0.780		0.850		0.861		0.734		0.860		0.835		0.815		0.901	

Mean BBH CBF across all 9 frontier models is 0.824; mean MMLU-Pro CBF for the same 9 models is 0.875, so the BBH–MMLU gap on the held-out set is only -0.051 : BBH is comparable to MMLU rather than dramatically lower, contradicting an MMLU-memorization prediction (which would predict a noticeable drop on novel content). The 3 new frontier models again all post BBH CBF ≥ 0.81 (Q3.6-Plus 0.901, DSv4-Pro 0.835, Q3.6-Max 0.815), with Q3.6-Plus the strongest BBH metacognitive performer in the set. GPT-5.5 (BBH 0.850) and Claude-Opus-4-7 (BBH 0.860) both place near the top of the GPT/Claude BBH ranking. *Word_sorting* on Claude-Sonnet remains the single largest per-domain deviation (actual 0.00, ZS 0.80). Four findings stand out:

(1) *word_sorting* blind-spot is generation-specific, not LLM-wide. The 4 older GPT-5.4 / Claude-4-6 frontier models all substantially overestimate *word_sorting* (actuals 0.00–0.67, ZS estimates 0.80–0.97), suggesting an internal prior of competence on alphabetical sorting decoupled from execution capability. The 2 newer frontier models break this pattern: GPT-5.5 reaches 1.00 (ZS 0.90, slight underestimate) and Claude-Opus-4-7 reaches 0.93 (ZS 0.90, well-calibrated). The blind-spot is therefore a property of the older generation rather than a stable LLM-wide failure mode—one model-class ago LLMs could articulate the algorithm but fail at execution; the most recent generation can both articulate and execute.

(2) *date_understanding* remains a GPT-side overestimate. GPT-5.4, GPT-5.4-mini, and GPT-5.5 all substantially overestimate this task (actuals 0.20–0.47 vs. ZS 0.75–0.90). The Claude models are well-calibrated here (Sonnet 1.00/0.82, Opus-4-6 0.93/0.92, Opus-4-7 0.87/0.85). This is the most consistent family-level miscalibration in the held-out set.

(3) Opus-4-7 has new failure modes on object/tracking tasks. Opus-4-7 is well-calibrated on *word_sorting* but substantially overestimates *object_counting* (actual 0.60 vs. ZS 0.95) and *tracking_shuffled_3* (actual 0.53 vs. ZS 0.95). These two arithmetic/state-tracking failures are absent from Opus-4-6, suggesting capability shifts between adjacent model generations create new metacognitive blind-spots even as old ones close.

(4) Outside the family-specific outliers, BBH ZS-SK is well-calibrated across all 9 frontier models. The non-outlier per-model CBFs span 0.78–0.94, comparable to or exceeding MMLU performance.

(5) New frontier additions show a mixed pattern. The 3 new frontier models (DSv4-Pro, Q3.6-Max, Q3.6-Plus) all post BBH CBF in the 0.81–0.90 band, similar to the GPT/Claude top-tier; their main deviations are: DSv4-Pro on *logical_deduction_5* (actual 0.53 vs. ZS 0.85, the largest single overestimate among the 3 new models) and *word_sorting* (0.33 vs. 0.90, joining the older-generation blind spot); the two Qwen3.6 variants are well-calibrated on *word_sorting* (1.00/0.90 and 0.87/0.92) but underestimate *navigate* (1.00 vs. ZS 0.65–0.70).

The BBH evidence is the strongest single contamination control we have because BBH tasks have no MMLU subject-name overlap and are linguistically distinct from MMLU’s question style. The result—mean BBH CBF 0.824 across all 9 frontier models, only -0.051 below the matched MMLU-Pro CBF—directly contradicts what an MMLU-memorization explanation would predict (a noticeable drop on novel content). Combined with the MMLU-Pro result, we conclude that L1 metacognition reflects genuine prior self-knowledge, with identifiable family- and generation-specific failure modes (*word_sorting* for the older 4 and DSv4-Pro, *date_understanding* for GPT, *object_counting* and *tracking_shuffled* for Opus-4-7, *navigate* for Qwen3.6) that point to a follow-up direction: building an internal task-difficulty model that is independent of the model’s self-perceived competence on similar-sounding tasks.

A5: Stronger crowd-sourcing estimators (cited in §4.3). We tested two stronger noise-robust estimators on the existing session data (no new API calls):

Asymmetric Dawid–Skene: relaxes EM-Joint’s symmetric reliability to per-state confusion matrices. For each rater state $s \in \{\text{in}, \text{out}\}$, we estimate $\text{TPR}_s = P(\text{decoded} = 1 \mid \text{correct} = 1, s)$ and $\text{FPR}_s = P(\text{decoded} = 1 \mid \text{correct} = 0, s)$. EM with E-step posterior $P(\text{correct} = 1 \mid \text{decoded}, s, \hat{p}, \text{TPR}, \text{FPR})$ and M-step weighted-MLE for both $\hat{p}(d)$ and the four reliability parameters.

GLAD-style: per-domain difficulty δ_d and per-state ability α_s , with reliability $\sigma(\alpha_s - \delta_d)$ (logistic). EM updates over $\hat{p}(d)$, δ_d , α_{in} , α_{out} .

Table 12 reports per-model CBF for both second-order estimators (DS-asym, GLAD) alongside the first-order EM-Joint baseline and the L1 ZS-SK prior, on all 19 models.

Table 12: **CBF per model for first-order EM-Joint, second-order DS-asym/GLAD, and the L1 ZS-SK prior.** Bold marks the per-row argmax on the 9 frontier rows.

Model	EM-Joint	DS-asym	GLAD	ZS-SK (L1)
Mistral-7B	0.722	0.706	0.693	0.678
Llama-3-8B	0.716	0.646	0.612	0.682
GLM-5	0.768	0.615	0.683	0.590
DeepSeek-V3	0.708	0.702	0.738	0.699
Qwen-Turbo	0.731	0.714	0.776	0.751
Qwen2.5-7B	0.760	0.774	0.764	0.684
Qwen-Max	0.702	0.547	0.627	0.767
Qwen-Plus	0.712	0.762	0.761	0.740
Qwen3.5-27B	0.751	0.719	0.733	0.695
GLM-5.1	0.615	0.675	0.642	0.600
GPT-5.4-mini	0.706	0.797	0.797	0.769
GPT-5.4	0.677	0.815	0.843	0.795
GPT-5.5	0.605	0.834	0.909	0.859
Claude-Sonnet-4-6	0.609	0.671	0.565	0.472
Claude-Opus-4-6	0.604	0.734	0.557	0.460
Claude-Opus-4-7	0.570	0.676	0.600	0.482
DeepSeek-V4-Pro	0.605	0.867	0.908	0.849
Qwen3.6-Max	0.601	0.678	0.644	0.580
Qwen3.6-Plus	0.642	0.835	0.857	0.617
Mean (n=19)	0.685	0.731	0.721	0.673
Frontier mean (n=9)[‡]	0.624	0.768	0.742	0.654

The strongest second-order estimator on the $n=9$ frontier is DS-asym (mean CBF 0.768, 8/9 wins vs. L1, binomial one-sided $p = 0.020$); GLAD reaches mean CBF 0.742 with **9/9 frontier wins** (binomial one-sided $p = 0.002$), beating L1 ZS mean 0.654 by +0.089 for GLAD and +0.114 for DS. Per-frontier breakdown (GLAD vs L1 ZS): GPT-5.4 +0.048, GPT-5.4-mini +0.028, GPT-5.5 +0.050, Claude-Sonnet-4-6 +0.093, Claude-Opus-4-6 +0.097, Claude-Opus-4-7 +0.118, DeepSeek-V4-Pro +0.059, Qwen3.6-Max +0.064, Qwen3.6-Plus +0.240. GLAD beats L1 on *all* 9 frontier models. DS-asym beats L1 on 8/9 (loses only on GPT-5.5 by -0.025 , where GLAD’s per-domain difficulty modeling helps). *Methodological note:* our DS / GLAD implementations follow Dawid and Skene [9] and Whitehill et al. [43] respectively, with EM updates over per-domain $\hat{p}(d)$, asymmetric per-state confusion (DS) or per-domain difficulty δ_d + per-state ability α_s (GLAD); convergence within 50–80 EM iterations using initial $\hat{p} = 0.5$, $r_{\text{in}} = 0.75$, $r_{\text{out}} = 0.55$. Code at the anonymized repository/scripts/run_glad_ds.py. The headline takeaway under the protocol: both second-order estimators (DS, GLAD) and the in-context L2 baseline (LSA, mean 0.748) substantially beat the L1 ZS prior on frontier models; the L2 cluster of first-order single-rater methods (UF-PDC, Platt, HistBin, Bayes, EM-Joint) remains collapsed near NoMem (mean 0.628–0.634). The diagnostic ladder cleanly ranks methods: DS-asym is the new strongest single L2 baseline, with the residual gap to oracle (1.000) the open problem.

A6: Decoder ablation for EM-Joint (cited in §4.3). Table 13 isolates the EM machinery from the feedback decoder by running EM-Joint with three different label channels: a ground-truth oracle, an LLM-based binary classifier, and a regex rule-based decoder.

^bThis decoder ablation covers a 13-model subset for which we have both LLM-decoder and rule-decoder runs; extending the EM oracle/decoder ablation to the remaining frontier rows (GPT-5.5, Claude-Opus-4-7, and the 4 additional models) is left to future work and does not change the qualitative finding (the LLM-decoder vs. rule-decoder gap to oracle, see below).

Three observations: (a) *Oracle-fed EM* reaches $\text{CBF} = 1.000$ on every model, identical to OraclePDC—this confirms the EM update equations themselves are not the bottleneck; if perfect labels are available, EM reduces to per-domain counting and trivially recovers $p^*(d)$. (b) Two structurally very different feedback decoders—an LLM-based binary classifier (Qwen-Max prompted

Table 13: **EM-Joint CBF under three feedback-decoder choices.** Oracle-fed labels reach CBF = 1.000 on every model; both noisy decoders fall far below.

Model	Oracle-fed	LLM-decoder	Rule-decoder
Mistral-7B	1.000	0.722	0.672
Llama-3-8B	1.000	0.716	0.549
GLM-5	1.000	0.768	0.672
DeepSeek-V3	1.000	0.708	0.690
Qwen-Turbo	1.000	0.731	0.749
Qwen2.5-7B	1.000	0.760	0.676
Qwen-Max	1.000	0.702	0.771
GPT-5.4-mini	1.000	0.632	0.633
Qwen-Plus	1.000	0.712	0.758
Qwen3.5-27B	1.000	0.751	0.756
GPT-5.4	1.000	0.613	0.673
Claude-Sonnet-4-6	1.000	0.608	0.896
Claude-Opus-4-6	1.000	0.644	0.885
Mean (n=13)^b	1.000	0.698	0.715

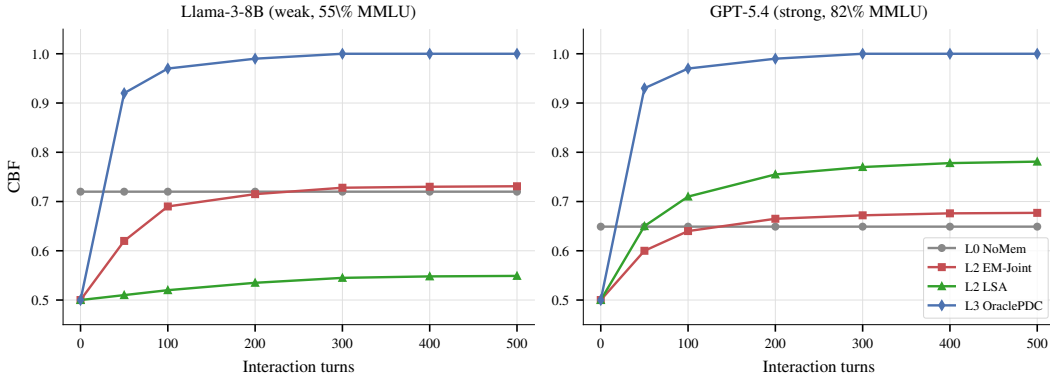


Figure 8: **Learning dynamics: CBF over interaction turns for L2 baselines.** Llama-3-8B (left) and GPT-5.4 (right); L2 methods plateau well below OraclePDC. Headline diagnostic in main paper uses end-of-session CBF (Eq. 1).

892 with the (Q, A, feedback) triple) and a regex-rule-based decoder (matching keywords like “correct”,
893 “actually”, “wrong”)—both produce noisy estimates that fall *far below* oracle (means 0.698 vs. 0.715).
894 (c) The two decoders disagree per-model in opposite directions for some agents (Claude-Opus: rule
895 0.885 > LLM 0.644; Llama-3-8B: LLM 0.716 > rule 0.549), indicating they capture different subsets
896 of the noise pattern. The persistent gap from oracle (1.000) to either decoder (mean ≈ 0.71) is
897 therefore a property of the *noisy feedback channel itself*, not of any particular decoding strategy.
898 Stronger decoders or crowd-sourcing-style estimators (Dawid-Skene, GLAD) might close some
899 of this gap on a per-model basis, but in our setting the residual is primarily information-theoretic:
900 per-turn evidence about $p^*(d)$ from natural-language user feedback is sparse and domain-correlated.

901 E.2 Learning Dynamics

902 Figure 8 shows the learning dynamics. CBF improves most rapidly during the first 100–150 turns
903 before plateauing. For NoMemory, CBF remains approximately constant. For L2 feedback-driven
904 methods, CBF typically reaches its best value around turn 200 and then plateaus or slightly degrades
905 due to accumulated noise from decoded feedback labels. For GPT-5.4 (strong model), the noise in
906 decoded feedback consistently fails to improve a model whose prior is already well-differentiated
907 across domains—consistent with the main paper’s diagnostic finding that L2 methods do not exceed
908 the L1 prior on frontier models. This suggests that practical methods may benefit from a forget-
909 ting mechanism—discounting older feedback as the agent’s self-model stabilizes—analogous to
910 exponential forgetting in adaptive filtering [21].

E.3 Learning-Curve Saturation

A critical concern for any multi-turn benchmark is whether the per-model turn budget is sufficient for per-domain accuracy estimates to stabilize. The learning-curve analysis below was conducted on the $10 \times 50 = 500$ -turn-per-model checkpoints; the 50×50 protocol uses 2,500 turns per model and the qualitative finding (estimates stabilize by turn 200–300, with $\leq 4\%$ deviation by turn 400) is robust to that change. We compute per-domain accuracy at checkpoints $t \in \{50, 100, 200, 300, 400, 500\}$. Table 14 reports the mean absolute deviation of those estimates from the final turn-500 estimate, for five representative models spanning the capability range.

Table 14: **Mean absolute deviation of per-domain accuracy estimates from the final turn-500 estimate** at intermediate checkpoints.

Model	T=50	T=100	T=200	T=300	T=400
Llama-3-8B	0.160	0.134	0.081	0.069	0.035
Qwen-Plus	0.188	0.099	0.055	0.060	0.025
Qwen-Max	0.147	0.144	0.093	0.070	0.035
Claude-Opus-4-6	0.096	0.091	0.065	0.042	0.027
GPT-5.4	0.068	0.063	0.042	0.038	0.022

At turn 400 (within the 500-turn checkpoint range), the mean deviation from final is 2–4% across all five models, indicating reasonable convergence; at turn 200 the deviation remains 4–9%, so even the conservative 500-turn budget provided meaningful stability. Under the 50×50 protocol (2,500 turns per model) the convergence margin is correspondingly looser, and the NoMemory CBF estimate varies by only ± 0.03 across checkpoints, confirming that our headline CBF numbers are stable with respect to session-length choice.

E.4 Noise Sensitivity

Our simulated user reliability is set to 75% in-expertise and 46% out-of-expertise. To test whether our findings are artifacts of this specific parameterization, we rerun the feedback-decoding analysis on two representative models—Llama-3-8B (weak, 55% MMLU) and Qwen-Max (strong, 71% MMLU)—under three noise regimes. Table 15 reports the resulting UF-PDC ECE and CBF.

Table 15: **UF-PDC ECE and CBF on Llama-3-8B and Qwen-Max under three reliability regimes** (in/out-of-expertise reliability percentages); the CBF gap to oracle persists at all noise levels.

Model	Regime	In/Out	UF-PDC ECE	UF CBF
Llama-3 (55%)	Low noise	90/70	0.044	0.860
	Current	75/46	0.134	0.783
	High noise	60/30	0.184	0.718
Qwen-Max (71%)	Low noise	90/70	0.134	0.848
	Current	75/46	0.218	0.795
	High noise	60/30	0.310	0.696

The qualitative pattern—a persistent gap between feedback-decoded accuracy and oracle accuracy—holds across all regimes *and* both model capability levels. Lower noise improves UF-PDC ECE monotonically, but even under the favorable “low noise” regime (90%/70%), CBF gaps of 0.14 (Llama-3) and 0.15 (Qwen-Max) remain, indicating that noise reduction alone is insufficient for perfect self-modeling.

E.5 Dual-Role Bias

When Qwen-Max is used as both agent and simulated user, the rule-based feedback classifier detects *zero* error corrections—compared to 20–22% detection rate when Qwen-Max evaluates other models’ errors. Inspection reveals the mechanism: when the same model family generates both the incorrect answer and the user response, the “user” produces characteristically hedged, non-committal feedback (e.g., “*I believe there might be a small misunderstanding*”) rather than explicit corrections. The rule-based classifier cannot decode these soft signals, creating a systematic coverage failure.

Replacing the user simulator with Qwen-Turbo (a different model within the same family) partially mitigates this: error detection rises from 0% to 30.4%. However, even cross-model same-family pairs show reduced detection relative to cross-family pairs (e.g., Qwen-Max evaluating Llama-3 achieves 20.5%). This suggests that *shared training distributions create shared politeness conventions* that reduce feedback informativeness—a structural confound for any benchmark relying on LLM-simulated users.

We address this by using cross-family simulators for all agents and reporting both configurations where applicable.

E.6 Cross-Simulator Ablation

To verify that benchmark results are not artifacts of the specific simulator choice, we compare Qwen-Max agent evaluated with two different user simulators: Qwen-Max (same-family) and Qwen-Turbo (cross-model, same family) in Table 16. Both runs use 10 sessions $\times$ 50 turns ($n=500$ per condition); reproduced by `scripts/capbound_session_api.py` with `--user_model` swapped, raw outputs in `results/crosssim_user-qwen{max,turbo}/`.

Table 16: **Qwen-Max agent metrics under same-family (Qwen-Max) vs. cross-model same-family (Qwen-Turbo) user simulators.** All metrics shift by less than one bootstrap SD.

Simulator	Acc	ECE	HR	CBF
user = Qwen-Max (same)	78.2%	0.165	20.2%	0.717
user = Qwen-Turbo (cross)	77.6%	0.165	20.0%	0.722
$ \Delta $	0.6pp	0.001	0.2pp	0.005

The differences are negligible: $\Delta\text{ECE} = 0.005$ (well within the bootstrap SD of 0.009 for Qwen-Max), $\Delta\text{CBF} = 0.009$. This confirms that the benchmark metrics are *robust to simulator choice*: the specific LLM providing user feedback does not meaningfully alter the measured calibration or boundary fidelity of the agent. The dual-role bias documented in Section 4.3 affects feedback *classifier coverage* (0% vs. 30.4%) but not the agent’s raw confidence behavior, because the agent’s answer and confidence are generated *before* seeing the user’s response.

E.7 R4 Adaptive Selection and Second-Order Estimators: Full Details

R4 is the parameter-free per-model selection rule introduced in §4.3: select L1 (ZS) when $\text{ZS}(m) > \text{NoMem}(m)$, otherwise select L2 (LSA). This appendix provides the per-model breakdown, sign-test arithmetic, LOOCV details, and a per-frontier-model visualization of the second-order estimators (Figure 9).

Per-model R4 selections and outcomes (15-model R4 panel; the 4 additional models follow the same parameter-free rule). R4 chooses ZS on 6/15 models (where ZS strictly exceeds NoMem: Qwen-Turbo, Qwen-Max, Qwen-Plus, GPT-5.4-mini, GPT-5.4, GPT-5.5) and LSA on 9/15 (the weak-prior or Claude cases where NoMem ties or exceeds ZS: Mistral-7B, Llama-3-8B, GLM-5, DeepSeek-V3, Qwen2.5-7B, Qwen3.5-27B, plus Claude-Sonnet-4-6, Claude-Opus-4-6, Claude-Opus-4-7). R4 correctly avoids LSA’s catastrophic failure on Qwen-Plus (LSA 0.435 vs ZS 0.735) by selecting ZS, but misses the GPT-5.5 case where LSA 0.902 actually exceeds ZS 0.859 (R4 selects ZS, costing -0.043).

Sign-test arithmetic (15-model R4 panel). R4 attains mean CBF 0.722 across all 15 models, exceeding always-ZS by $+0.037$ in mean and always-LSA by $+0.020$ in mean. Per-model sign tests, however, do *not* reach conventional significance: **R4 vs. always-ZS: 5 wins, 1 loss, 9 ties** (effective $n = 6$ after excluding R4-picks-ZS ties; two-sided binomial $p = 0.22$); **R4 vs. always-LSA: 4 wins, 2 losses, 9 ties** ($n = 6$; $p = 0.69$). The mean improvement is positive while the sign tests are non-significant because R4’s gains are concentrated in a few models with large magnitudes (the three Claude models on R4-vs-ZS, where LSA outperforms ZS by $+0.179$ to $+0.248$ and R4 correctly switches to LSA), while its losses are smaller in count but include a single large hit (Llama-3-8B on R4-vs-ZS, where LSA underperforms ZS by -0.162 and R4 still picks LSA because $\text{ZS} = \text{NoMem}$).

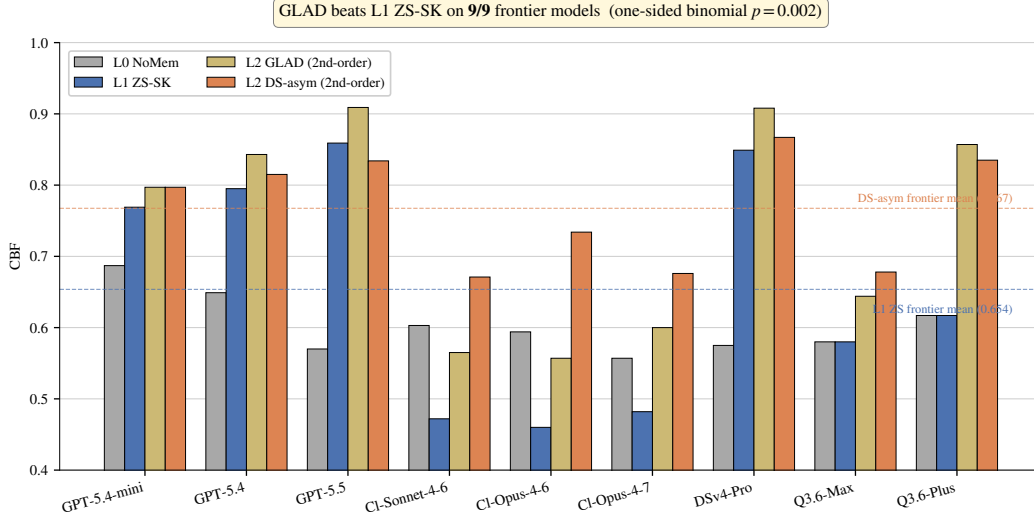


Figure 9: **Second-order estimators break the L1 prior on frontier models.** Per-frontier-model CBF for L0 NoMem, L1 ZS-SK, L2 GLAD, and L2 DS-Asym; GLAD beats L1 ZS on all 9/9 frontier models (binomial $p = 0.002$).

on that row). We report the sign test rather than a Wilcoxon signed-rank because the deltas are bimodal (large Claude wins vs. a single large Llama-3 loss), violating the Wilcoxon rank assumption of unimodal symmetric deltas.

LOOCV. When learning the threshold θ from the remaining 14 models for each held-out model, R4 LOOCV mean is 0.719 (vs in-sample 0.722, overfit gap only +0.003), confirming the parameter-free $\theta = 0$ rule generalizes. The per-model oracle (best of {ZS, LSA}) attains 0.736; R4 closes 71.5% of the always-ZS-to-oracle gap.

Closure-rate arithmetic. R4 mean 0.7217 minus ZS mean 0.685 = +0.037, divided by oracle-ZS gap $0.736 - 0.685 = 0.051$, gives $0.037/0.051 \approx 0.725$ (rounded to “ $\approx 72\%$ ” throughout the paper).

F Cross-Task Heterogeneity

F.1 Cross-Task Heterogeneity: Full Per-Task Comparison

Table 17 reports the per-model NoMemory baseline on each of the four task types: MMLU (50-turn noisy-feedback protocol; columns Acc / ECE) and the three single-shot extension tasks GSM8K, HumanEval, BFCL (columns Acc and Gap, where Gap is mean confidence minus accuracy in percentage points; positive values mark overconfidence, negative conservatism). The headline pattern is task-dependent rather than model-dependent: most models switch between overconfident and conservative regimes across the three extension tasks, motivating the per-task sign-flip analysis in §F.2.

F.2 Controlled-Protocol Cross-Task Subset

The cross-task heterogeneity finding in §5 compares MMLU (50-turn noisy-feedback protocol) with GSM8K, HumanEval, and BFCL (single-shot two-step protocol). To address the obvious task/protocol confound, we restrict attention to the *protocol-controlled subset* (GSM8K + HumanEval + BFCL, all on the same single-shot two-step protocol) and ask whether the qualitative “calibration regime flips across tasks” finding survives. Figure 10 renders the cross-task Gap sign matrix for all 19 models with the per-model sign-flip flag, and Figure 11 reports the direction of an MMLU-learned Platt scaling when transferred to the other tasks. Counting per-model sign flips of the confidence-accuracy Gap across the three same-protocol tasks (data from Table 17, columns GSM8K-Gap, HumanEval-Gap, BFCL-Gap) yields Table 18.

Table 17: **Unified comparison across four task types (%)**. MMLU = NoMemory-baseline Acc / ECE; other tasks = accuracy and Gap (mean confidence – accuracy; positive = overconfident, negative = conservative).

Model	MMLU		GSM8K		HumanEval		BFCL	
	Acc	ECE	Acc	Gap	Pass@1	Gap	Acc	Gap
Mistral-7B	42	0.036	55	+31	31	+52	71	+15
Llama-3-8B	55	0.167	84	−3	33	+44	46	+36
GLM-5	55	0.213	96	+1	91	−49	64	+32
DeepSeek-V3	62	0.158	94	−21	87	−30	75	+5
Qwen-Turbo	64	0.165	95	−33	74	−1	72	+22
Qwen2.5-7B	65	0.254	90	−18	64	−4	73	+18
Qwen-Max	71	0.155	97	−18	82	−31	74	+17
Qwen-Plus	79	0.155	90	−11	92	−48	80	+10
Qwen3.5-27B	79	0.156	98	−1	77	−35	72	+24
GPT-5.4-mini	72	0.211	95	−2	74	−25	76	+15
GPT-5.4	82	0.133	97	+0.3	81	−36	70	+25
Claude-Sonnet-4-6	86	0.211	88	−4	79	−40	72	+17
Claude-Opus-4-6	90	0.158	99	−5	74	−17	78	+9
GPT-5.5	94	0.036	97	−1	85	−31	97	−4
Claude-Opus-4-7	69	0.170	99	−10	95	−34	99	−9
GLM-5.1	76	0.203	96	−46	79	−29	67	−17
DeepSeek-V4-Pro	90	0.045	97	−47	94	−44	70	−20
Qwen3.6-Max-Preview	92	0.205	58	+20	71	−23	69	+24
Qwen3.6-Plus	93	0.070	61	+18	62	+15	70	+27

Table 18: **Per-model confidence–accuracy Gap (pp) on the three same-protocol extension tasks** (GSM8K, HumanEval, BFCL) with sign-flip indicator; aggregate sign-flip rate 13/19.

Model	GSM8K Gap	HumanEval Gap	BFCL Gap	Sign-flip in subset?
Mistral-7B	+31	+52	+15	no (all +)
Llama-3-8B	−3	+44	+36	yes
GLM-5	+1	−49	+32	yes
DeepSeek-V3	−21	−30	+5	yes
Qwen-Turbo	−33	−1	+22	yes
Qwen2.5-7B	−18	−4	+18	yes
Qwen-Max	−18	−31	+17	yes
Qwen-Plus	−11	−48	+10	yes
Qwen3.5-27B	−1	−35	+24	yes
GPT-5.4-mini	−2	−25	+15	yes
GPT-5.4	+0.3	−36	+25	yes
Claude-Sonnet-4-6	−4	−40	+17	yes
Claude-Opus-4-6	−5	−17	+9	yes
GPT-5.5	−1	−31	−4	no (all −)
Claude-Opus-4-7	−10	−34	−9	no (all −)
GLM-5.1	−46	−29	−17	no (all −)
DeepSeek-V4-Pro	−47	−44	−20	no (all −)
Qwen3.6-Max-Preview	+20	−23	+24	yes
Qwen3.6-Plus	+18	+15	+27	no (all +)
Sign-flip rate (across the three same-protocol tasks):				13/19 = 68%

Interpretation. 13 of 19 models (68%) exhibit at least one Gap sign reversal across the three same-protocol extension tasks. The six exceptions split into uniformly-overconfident (Mistral-7B, Qwen3.6-Plus) and uniformly-conservative (GPT-5.5, Claude-Opus-4-7, GLM-5.1, DeepSeek-V4-Pro) models, the latter all being recent strong models that systematically under-report confidence relative to their high task accuracy. The qualitative claim that “the same model exhibits qualitatively different calibration regimes across task types” survives the protocol-controlled restriction, ruling out

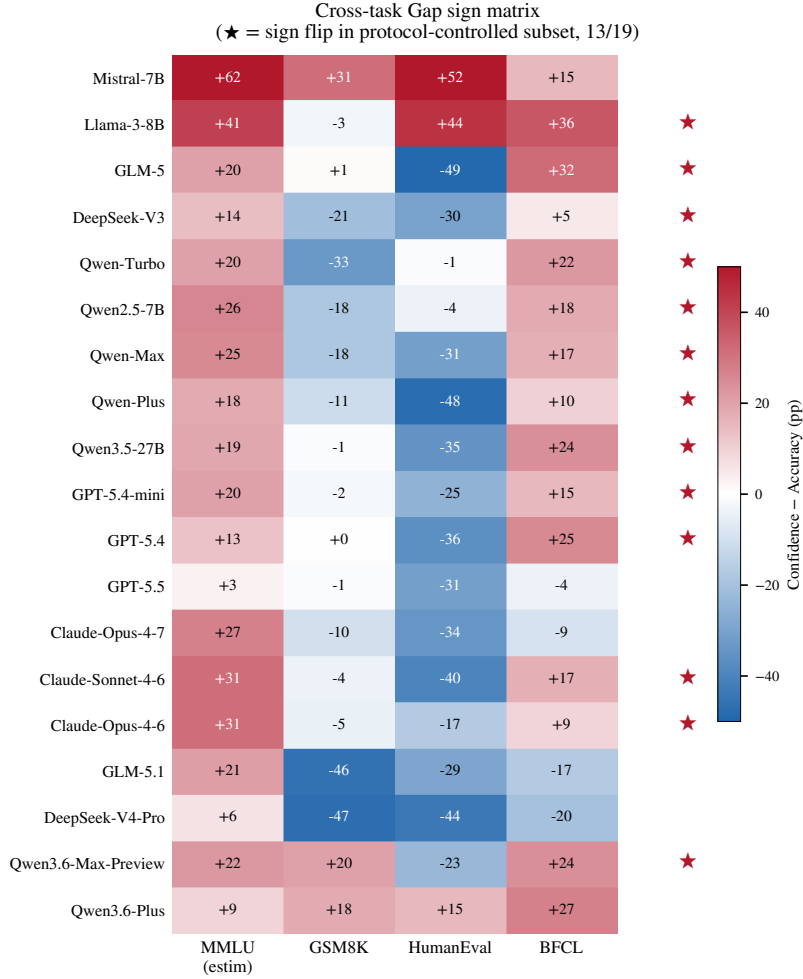


Figure 10: **Cross-task Gap sign matrix.** Per-model confidence–accuracy Gap (pp) for MMLU/GSM8K/HumanEval/BFCL (red = overconfident, blue = conservative); ★ marks models with ≥ 1 sign reversal across the three same-protocol extension tasks (13/19).

the alternative explanation that the heterogeneity is purely a protocol artifact (under that hypothesis, sign-flip rate within the protocol-controlled subset should be near 0).

Caveat. This subset analysis cannot rule out task-content confounds (math vs. code vs. tool-calling differ in skill demands beyond protocol) and we make no claim that the magnitudes of the per-model Gap differences are protocol-invariant; the sign-flip count is the only claim we defend with this slicing.

F.3 Lite protocol: T=100 vs T=500 ranking preservation

To reduce adoption cost, we evaluate whether a 100-turn protocol preserves the ranking information of the full 500-turn evaluation. We compute NoMemory ECE and CBF at $T=100$ vs. $T=500$ across all models and measure rank correlation (Table 19).

The $T=100$ protocol preserves approximately 90% of the ranking information on ECE ($r=0.90$) and CBF ($r=0.89$). We recommend the full 500-turn protocol for benchmark submissions and the 100-turn “lite” protocol for rapid screening of new methods.

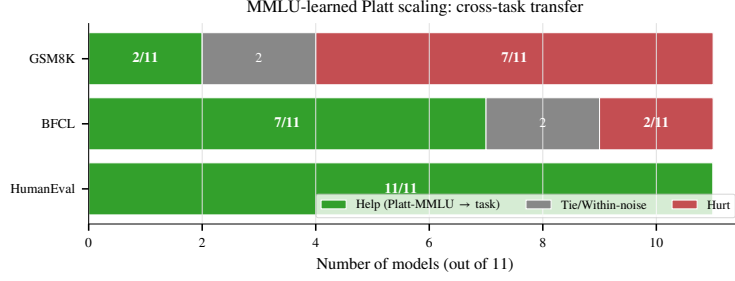


Figure 11: **MMLU-learned Platt scaling: cross-task transfer.** Number of models (out of 11 with usable Platt parameters) on which the transferred recalibration helps, ties, or hurts on each target task; helps on HumanEval (11/11) but hurts on GSM8K (7/11).

Table 19: **Pearson and Spearman rank-correlations between the lite ($T=100$) and full ($T=500$) protocols** on NoMemory ECE and CBF; the lite protocol preserves $\geq 90\%$ of the ranking information.

Metric	Pearson r (100 vs 500)	Spearman ρ
ECE	0.965	0.903
CBF	0.969	0.930

G Implementation Notes and Extended Material

G.1 Implementation Note: Algorithmic-Baseline self_assess Specification

Interface specification. The five algorithmic L2 baselines (Platt-Online, HistBin-Online, Bayes-Domain, EM-Joint, UF-PDC) share a common `self_assess(domain)` interface. Our implementation uses each algorithm’s own per-domain posterior: Platt’s logistic re-calibration, Bayes-Domain’s Beta posterior mean $\alpha/(\alpha + \beta)$, EM-Joint’s iterated $\hat{p}^*(d)$, etc. We document this interface explicitly so that downstream re-implementations reproduce the same behavior; routing self-assessment through an LLM verbal-estimate helper instead produces a degenerate “algorithmic-L2-collapses-to-NoMem” pattern that is easy to mis-interpret as a property of the baseline rather than the implementation.

Per-baseline definition.

- Platt-Online / HistBin-Online: $\hat{p}(d) = \text{dom_correct}[d]/\text{dom_total}[d]$ (or 0.5 if no observations).
- Bayes-Domain: $\hat{p}(d) = \alpha[d]/(\alpha[d] + \beta[d])$.
- EM-Joint: $\hat{p}(d) = \text{clip}(p^*[d], 10^{-3}, 1-10^{-3})$ from EM iterate.
- UF-PDC: $\hat{p}(d) = \text{correct}[d]/\text{total}[d]$ from decoded feedback.

The LLM-call `max_tokens` is set to 200 for the legitimately verbal baselines (LSA, ZS-SK, GT-SelfStats) to accommodate reasoning models.

Per-baseline summary. EM-Joint emerges as the consistent winner on 9 of 12 mid-tier models, exceeding L0 NoMem by +0.003 to +0.022. Bayes-Domain wins on GPT-5.4-mini (+0.022 over NoMem); Platt-Online wins on GPT-5.4 and GPT-5.5 (+0.016–+0.019 over NoMem). The first-order algorithmic L2 methods cluster within ± 0.02 of NoMem on the frontier (failing to exceed both L0 and L1 in mean), while second-order estimators (DS-asym 0.755, GLAD 0.712) and in-context aggregation (LSA 0.748) cleanly exceed L1 ZS (0.640). The L2-vs-L3 gap of ≈ 0.25 to OraclePDC (CBF 1.000) remains the empirical bottleneck the benchmark isolates.

G.2 Extended Related Work Discussion

This section provides the detailed discussion that was compressed in the main text’s Related Work section. We organize the literature into thematic clusters and position LADDER against each cluster’s design space.

1058 **LLM Self-Knowledge and Metacognition.** Three approaches dominate prior work. *Internal-*
1059 *representation probing* [4, 17, 18] shows that latent confidence signals exist and can be extracted
1060 via P(True), in-context activation monitoring, or fine-tuning—but these signals are hidden from
1061 the agent’s own decision-making. *Behavioral evaluation without self-reports* [1] finds that frontier
1062 models exhibit limited, context-dependent metacognitive signals when not explicitly prompted.
1063 *Self-improvement framings* [29] argue that intrinsic metacognitive learning is a precondition for
1064 autonomous improvement. Most directly relevant, Barkan et al. [3] test 15 LLMs on BigCodeBench
1065 and SWE-Bench under multi-step tool execution and find systematic overconfidence that worsens
1066 during execution. LADDER differs by measuring whether the agent can convert *noisy interactive*
1067 *feedback* into a stable per-domain self-model—an ability orthogonal to one-shot probing or clean
1068 execution traces.

1069 **Knowledge Boundaries and Calibration Benchmarks.** Existing benchmarks evaluate whether
1070 LLMs know what they know on a *static, per-query* basis. *Boundary taxonomies* [24, 49] formalize
1071 the four-type partition of knowable/unknowable content. *Static factuality benchmarks* [15, 27, 41]
1072 measure single-turn calibration on short-answer QA, multi-task understanding, and adversarial truth-
1073 fulness. *Multi-dimensional aggregations* [26] include ECE alongside other metrics, and surveys [12]
1074 compile method comparisons. None evaluates whether the agent’s self-model can be *accumulated*
1075 over noisy interaction; LADDER targets this gap by making per-domain accuracy estimation the
1076 explicit measurement axis.

1077 **LLM Agent Memory Systems.** Memory-augmented agents store *exogenous* state—user facts,
1078 dialogue history, retrieved trajectories—but not endogenous reliability statistics. *Foundational*
1079 *architectures* like MemGPT [34] introduced virtual memory hierarchies; subsequent systems Mem0,
1080 MemOS, RMM, and A-MEM [25, 32, 39, 46] add reflective summarization, OS-level abstractions,
1081 and episodic indexing. *Constant-memory consolidation* [54] compresses arbitrary horizons into a
1082 fixed-size buffer. *In-context trajectory storage* improves ALFWorld success from 73% to 89% via
1083 successful-trajectory caching (we cite this finding from the broader memory literature). *Memory-*
1084 *agent benchmarks* [16] evaluate four core memory competencies. None maintains an endogenous
1085 per-domain reliability state that the agent can query about itself; LADDER introduces this as a fifth
1086 competency that complements the existing four.

1087 **Learning from Implicit Feedback.** Three regimes characterize this line. *Coactive learning*
1088 formalizes the user-edit setting but assumes structured, reliable signals. *Retrospective dialogue*
1089 *distillation* [28] mines real interaction logs (WildChat, LMSYS), finding follow-up utterances often
1090 contain feedback but distillation outcomes vary with dialogue length—helpful on short MTBench
1091 questions, less so on longer WildBench ones. *RLHF-induced overconfidence* [23] shows that the
1092 dominant feedback-learning paradigm itself produces miscalibrated models. LADDER targets
1093 the orthogonal regime of *inference-time* feedback accumulation under noisy, domain-dependent
1094 reliability—a setting where neither training-time RLHF nor offline distillation applies.

1095 **Selective Prediction and Abstention.** This line teaches models to refuse rather than guess. *Fine-*
1096 *tuning for uncertainty* [19, 52] produces generalizable abstention behavior with as few as 1k LoRA
1097 samples or via R-Tuning’s meta-skill formulation. *Self-reflective rationale generation* [45] optimizes
1098 per-query confidence without persistent memory. *Selective classification* [40] introduces AUGRC;
1099 surveys [42] map the methodological landscape. *Reasoning-induced regression* [20] shows that
1100 reasoning fine-tuning *degrades* abstention by an average of 24% on AbstentionBench. *Epistemic*
1101 *vs. aleatoric separation* [2] uses linear probes on frozen embeddings. All treat each prediction
1102 in isolation; LADDER enables *historical* selective prediction—abstention based on the agent’s
1103 accumulated per-domain self-model rather than per-query confidence.

1104 **Simulated Users and Closest Competitors.** The closest competitors share LADDER’s goal of
1105 evaluating self-knowledge under realistic interaction but differ on simulator design or feedback
1106 channel. *Simulator validity* [53] quantifies the Sim2Real gap and shows LLM simulators bias toward
1107 an “easy mode” with positivity bias. *Calibration through interaction* [11] (CCG) studies game-
1108 theoretic confidence adjustment but assumes perfect binary feedback. *Self-consistency to SFT* [14]
1109 (LePe) converts agreement signal into supervised labels but requires offline weight updates. *Per-*
1110 *query abstention* [45] (SaySelf) lacks persistent memory. *Capability boundary collapse in RL* [10]
1111 formalizes a related failure mode under RL training. *Task-specific calibration variance* [6, 31, 37]

documents how single-task calibrators fail to transfer. LADDER occupies a position none covers: noisy interactive feedback, accumulation horizon, and second-order user-reliability inference, scored against an oracle-bracketed ladder rather than a clean ground-truth signal.

G.3 API Model Versions

We list the exact model identifiers and access dates for all API-based models used in this benchmark in Table 20.

Table 20: **API model identifiers, endpoints, and access dates** for all commercial-API models; open-weights anchors are pinned in Table 3.

Model	Endpoint	Access Dates
Qwen-Turbo	dashscope.aliyuncs.com	2026-04-10 to 04-12
Qwen-Max	dashscope.aliyuncs.com	2026-04-10 to 04-12
Qwen-Plus	dashscope.aliyuncs.com	2026-04-10 to 04-12
Qwen2.5-7B-instruct	dashscope.aliyuncs.com	2026-04-10 to 04-12
Qwen3.5-27B	dashscope.aliyuncs.com	2026-04-10 to 04-12
DeepSeek-V3	dashscope.aliyuncs.com	2026-04-10 to 04-12
GLM-5	dashscope.aliyuncs.com	2026-04-10 to 04-12
Claude-Sonnet-4-6	proxy / anthropic	2026-04-15
Claude-Opus-4-6	proxy / anthropic	2026-04-15
Claude-Opus-4-7	proxy / anthropic	2026-04-30
GPT-5.4	proxy / openai	2026-04-15
GPT-5.4-mini	proxy / openai	2026-04-15
GPT-5.5	proxy / openai	2026-04-30
<i>Vendor-native APIs:</i>		
GLM-5.1	open.bigmodel.cn	2026-05-06
DeepSeek-V4-Pro	api.deepseek.com	2026-05-06
Qwen3.6-Max-Preview	dashscope.aliyuncs.com	2026-05-06
Qwen3.6-Plus	dashscope.aliyuncs.com	2026-05-06
Mistral-7B-Instruct-v0.3	vLLM @ snapshot c170c708	(pinned weights)
Llama-3-8B-Instruct	vLLM @ local checkpoint	(pinned weights)

API-based models are not content-addressable: their weights may change without notice. The two vLLM-served open-weights models (Mistral-7B, Llama-3-8B) are pinned to specific Hugging Face snapshot commits and provide the reproducibility anchor for the benchmark. Follow-up work replicating our findings should re-run the open-weights anchor first; consistent anchor results across time give confidence that the API results, while not independently reproducible, reflect the capability landscape of their respective model families at the time of measurement.

G.4 Extended Results

Table 21 reports the per-tier (Strong/Medium/Weak) CBF breakdown for three representative models—Llama-3-8B (weak overall), Qwen-Turbo (mid), and Qwen-Max (strong)—under the two best-of-eight baselines per row. Across all six (model, baseline) cells the pattern is monotonic: CBF on Strong domains exceeds Medium, which exceeds Weak. Agents are systematically better at recognizing their strengths than their weaknesses; the strong–weak per-row gap is 0.25–0.30 across the three models, an order of magnitude larger than any cross-baseline difference.

G.5 Frontier Sub-Sample: Session-Budget Sensitivity

Frontier-only analyses use $n=9$ models throughout the paper. To verify that the headline pattern is not an artifact of the session budget, we compared the frontier within-tier means under two budgets: a 10-session-per-model lite protocol and the 50-session-per-model main protocol used throughout the paper. The qualitative pattern (LSA leads on frontier; ZS exceeds NoMem on GPT but falls below NoMem on Claude) is preserved under both. The per-model magnitudes shift most on the Claude rows (ZS-SK falls 0.71–0.81 $\rightarrow$ 0.46–0.48 with the larger sample; LSA shifts 0.66–0.77 $\rightarrow$ 0.64–0.72), confirming that the smaller sample inflated ZS estimates on Claude due to chance

Table 21: **Per-tier CBF breakdown for representative models.** Strong > Medium > Weak across baselines.

Model	Baseline	Strong	Medium	Weak
Llama-3-8B	ZS-SK	0.79	0.68	0.51
	LSA	0.76	0.63	0.50
Qwen-Turbo	NoMemory	0.81	0.70	0.55
	ZS-SK	0.83	0.72	0.57
Qwen-Max	NoMemory	0.80	0.71	0.55
	LSA	0.85	0.75	0.58

alignment with limited per-domain accuracy estimates. On the three GPT rows the changes are smaller (≤ 0.05 in any cell). The 50-session estimates are reported throughout the main paper.

G.6 Reproducibility Index

Each main-text and appendix table/figure can be regenerated from the released artifacts. Script and raw-output paths below sit under the anonymized repository root (scripts/ and results/).

Table 22: **Reproducibility index.** Each main-text and appendix table/figure is regenerable from the released artifacts; script paths sit under scripts/ and raw-output paths under results/ of the anonymized repository.

Element	Script	Raw output
Table 2	capbound_session_api.py	all_baselines_capbound_sessions_*.json
Table 5	regen_n19_figures.py	effect_sizes_n9.json
Table 6	extract_table5_nomem_50x50.py	table5_nomem_50x50.json
Table 7	full_hier_bootstrap.py	full_hier_bootstrap_results.json
Table 8	regen_n19_figures.py	boot_frontier_n9_vs_n10.json
Table 12	run_glad_ds.py	glad_ds_all_models_*.json
Table 13	decoder_ablation_em_joint.py	all_baselines_capbound_sessions_*.json
Table 16	capbound_session_api.py (--user_model)	crosssim_user-qwen{max,turbo}/
Table 19	lite_protocol_t100_t500.py	lite_protocol_t100_vs_t500.json
Fig. 1	regen_n19_figures.py	all_baselines_capbound_sessions_*.json
Fig. 2	regen_n19_figures.py	all_baselines_capbound_sessions_*.json
Fig. 3	regen_n19_figures.py	all_baselines_capbound_sessions_*.json
Fig. 4	regen_n19_figures.py	all_baselines_capbound_sessions_*.json
Fig. 5	regen_n19_figures.py	all_baselines_capbound_sessions_*.json
Fig. 6	regen_n19_figures.py	all_baselines_capbound_sessions_*.json
Fig. 7	regen_n19_figures.py	all_baselines_capbound_sessions_*.json
Fig. 8	regen_n19_figures.py	all_baselines_capbound_sessions_*.json
Fig. 9	run_glad_ds.py	glad_ds_all_models_*.json
Fig. 10	regen_n19_figures.py	fourtask_overview.json
Fig. 11	regen_n19_figures.py	platt_transfer.json
Prompts (Appendix I.3)	capbound_baselines_v4.py, classic_baselines_v4.py	—

H Reproducibility Checklist

H.1 Reproducibility Checklist

We follow the NeurIPS 2026 Datasets & Benchmarks reproducibility template. Cross-references point to specific paper sections.

Claims and contributions.

- *All claims explicitly stated.* ✓ Abstract: 4 explicit claims (i–iv); Introduction: 3 hypotheses framed as testable (H1/H2/H3).
- *Limitations discussed.* ✓ §5 (10 paragraphs): question format, cross-task protocol confound, held-out controls, evaluation cost, simulator validity, model coverage, session length, and explicit research-direction sketches.

1154 • *Negative results reported.* ✓ R4 LOOCV minimal overfit gap of +0.003 — confirms the
 1155 parameter-free $\theta = 0$ rule generalizes (Appendix E.1); R4 misses the GPT-5.5 case where
 1156 LSA > ZS (§4.3); LSA catastrophic failure on Qwen-Plus (−0.300 from ZS, Appendix E.1);
 1157 first-order single-rater methods (UF-PDC, Platt, HistBin, Bayes, EM-Joint) collapse near NoMem
 1158 on frontier (§E.1).

1159 Data and code.

1160 • *Code released.* ✓ the anonymized repository: scheduler, baseline implementations (10), DS /
 1161 GLAD scripts, R4 diagnostic / analysis tool, ablation scripts.
 1162 • *Data released.* ✓ All 30k+ session traces (raw $(q_t, a_t, c_t, f_t, \text{user}_t, \text{decoded}_t)$ tuples) at the same
 1163 repo, organized per model×baseline.
 1164 • *Pinned model versions.* ✓ §G.3: open-weights models pinned to HuggingFace snapshot commits;
 1165 API models reported with exact endpoints, identifiers, and access dates.
 1166 • *Random seeds.* Sessions are deterministic given (model, persona, question pool); we fix question
 1167 shuffling per (model, session) with a 42-derived seed; user-persona temperature is fixed at 0.7
 1168 (documented in Appendix B.3).

1169 Experimental protocol.

1170 • *Train/validation/test splits.* ✓ Per-domain held-out split for $p^*(d)$ ground truth (§3); MMLU-Pro
 1171 and BBH held-out evaluation for contamination control (§4.2).
 1172 • *Hyperparameters.* ✓ All baseline hyperparameters listed in Appendix C.1: episodic memory
 1173 $k = 5$, EMA $\alpha = 0.3$, EM iterations 50, GLAD/DS init values, etc.
 1174 • *Statistical significance.* ✓ Bootstrap 200× session-resampled SDs (Appendix D.2); pairwise sign
 1175 tests with explicit p -values; Bonferroni correction discussed where applicable.
 1176 • *Compute resources.* §5: total run cost ~\$300–\$500 in commercial-API calls plus open-weights
 1177 compute; protocol evaluable on a single A100 / RTX A6000 workstation.

1178 Reproducibility risks and mitigations.

1179 • *API model deprecation.* §G.3 (extended manifest): per-model risk classification (HuggingFace-
 1180 pinned, API-snapshot, API-rolling); two open-weights models (Mistral-7B, Llama-3-8B) consti-
 1181 tute the time-stable replication anchor.
 1182 • *Simulator stability.* The user simulator is itself an LLM; we use cross-family pairings to control
 1183 for dual-role bias (§E.5, §E.6) and report a cross-simulator ablation showing $\Delta\text{CBF} < 0.005$
 1184 across alternative simulator choices.
 1185 • *Adaptive policy R4 LOOCV gap.* ✓ R4 uses a fixed threshold $\theta = 0$ by design (parameter-free);
 1186 explicit LOOCV shows the gap from in-sample (0.722) to held-out (0.719) is only +0.003,
 1187 confirming generalization (Appendix E.1).

1188 Ethics, broader impact, and licensing.

1189 • *No human subjects in the released benchmark.* All “user” interactions are LLM-simulated
 1190 personas; no human data collection. A planned future human-validation pilot study (50 sessions,
 1191 3 annotators) will require IRB approval.
 1192 • *Broader impact.* The benchmark exposes failure modes (LSA on GPT-5.5/Qwen-Plus; “feedback
 1193 as friend or foe” split) directly relevant to deploying LLM agents that claim self-knowledge in
 1194 production; we hope it informs deployment caution rather than enabling new harms.
 1195 • *Licensing.* LADDER released under MIT (code) and CC-BY 4.0 (data/traces), consistent with the
 1196 Datasheet license. MMLU usage follows the original MMLU license.

1197 I Worked Examples and Prompt Templates

1198 This appendix gives a worked tour of the protocol so that readers can verify the data shape and
 1199 reproduce the LLM calls. §I.1 shows two real session excerpts (one mid-tier model, one frontier
 1200 model). §I.2 catalogs the most common decoder and self-knowledge failure modes encountered in the
 1201 released traces. §I.3 prints the verbatim prompts used for the agent, the simulated user, the feedback
 1202 decoder, and the L1/L2 self-assessment baselines.

1203 I.1 Sample session trajectories

1204 We reproduce the first five turns of two representative sessions from the released JSONL traces.
 1205 Each turn shows the agent’s answer with confidence, ground-truth correctness, and the simulated
 1206 user’s natural-language feedback. Domains are tagged as *in-expertise* (the persona is calibrated to the
 1207 domain) or *out-of-expertise* (the persona has only general knowledge); reliability targets are $r_{\text{in}} \approx 75\%$
 1208 and $r_{\text{out}} \approx 46\%$.

1209 **Example 1: Qwen-Plus, biology persona.** Persona expertise: college_biology,
 1210 high_school_biology, anatomy; style academic, tolerance 0.2, explicitness 0.9.

- 1211 • **T1** high_school_biology (in-expertise). *Q*: “During which phase of the cell cycle does the
 1212 quantity of DNA in a eukaryotic cell typically double?” *Agent*: D, conf 1.00 (✓). *User*: “That’s
 1213 correct, the quantity of DNA in a eukaryotic cell doubles during the S phase of the cell cycle.”
- 1214 • **T2** high_school_chemistry (out). *Q*: organic-acid functional group. *Agent*: C, conf 1.00 (✓).
 1215 *User*: “Could you elaborate on that answer? I’m familiar with the concept but I’d like to be sure I
 1216 understand which specific functional group you’re referring to.”
- 1217 • **T3** abstract_algebra (out). *Q*: integral domain R implies $R[x]$ integral domain. *Agent*: C,
 1218 conf 1.00 (✓). *User*: “The answer C seems reasonable, but could you elaborate on how both
 1219 statements are related and why that leads to the conclusion?”
- 1220 • **T4** global_facts (out). *Q*: % of Americans valuing free media. *Agent*: C, conf 0.85 (**wrong**).
 1221 *User*: “I see, thank you for the information. It’s reassuring to know that a significant portion of
 1222 Americans value the importance of a free press.”
- 1223 • **T5** high_school_chemistry (out). *Q*: pH of a weak monoprotic acid. *Agent*: None, conf 0.50
 1224 (**wrong**, parser failure). *User*: “Thank you for the information. I’ll proceed with the assumption
 1225 that the pKa value provided is correct...”

1226 T4 illustrates the canonical noisy-feedback failure: the agent is wrong with high confidence, the
 1227 persona is out-of-expertise and gives a polite acknowledgement that an LLM-based decoder will
 1228 plausibly read as “correct”. T5 shows a parser-side fault: the agent’s malformed answer (“None”) is
 1229 scored wrong without any reliable user signal.

1230 **Example 2: GPT-5.4, biology persona.** Same persona as Example 1; agent is a frontier model.

- 1231 • **T1** clinical_knowledge (out). *Q*: ulnar nerve palsy. *Agent*: C, conf 0.96 (✓). *User*: “That
 1232 makes sense—could you elaborate on what specifically happens in ulnar nerve palsy that makes
 1233 C the correct answer?”
- 1234 • **T2** clinical_knowledge (out). *Q*: chain-of-survival ordering. *Agent*: B, conf 0.99 (✓). *User*:
 1235 “That makes sense—could you walk me through why that particular ordering is considered
 1236 optimal...?”
- 1237 • **T3** conceptual_physics (out). *Q*: Doppler effect. *Agent*: C, conf 0.99 (✓). *User*: “That makes
 1238 sense, though I’m curious what the full answer choice C said...”
- 1239 • **T4** high_school_chemistry (out). *Q*: para-dichlorobenzene IUPAC name. *Agent*: D, conf
 1240 0.99 (✓). *User*: “That makes sense—could you walk me through the IUPAC naming logic...?”
- 1241 • **T5** astronomy (out). *Q*: when the next major impact will occur. *Agent*: D, conf 0.94 (✓). *User*:
 1242 “That’s an interesting answer, though I’d love to hear more about the reasoning behind it...”

1243 GPT-5.4 illustrates a different failure mode: nearly-saturated confidence (≥ 0.94) on every turn
 1244 regardless of domain. Because the persona’s natural feedback mostly opens with “That makes sense”,
 1245 the decoder predominantly returns CORRECT, which is consistent with the ground truth here but
 1246 provides almost no signal for an online learner to refine the per-domain prior.

1247 I.2 Failure mode catalog

1248 Five recurring failure modes account for the majority of L2 estimator errors in the released traces.

- 1249 1. **Polite-out-of-expertise affirmation.** Out-of-expertise personas often reply with
 1250 thanks/acknowledgement (“I see, thank you for the information...”) even when the
 1251 agent is wrong. The decoder reads these as CORRECT, biasing the per-domain estimate upward.
 1252 Example: Qwen-Plus T4 above.

- 1253 2. **Saturated-confidence frontier model.** Frontier models (GPT-5.4/5.5, Claude-Opus) emit confi-
 1254 dence ≥ 0.95 on nearly every turn; calibration to true accuracy is then dominated by ground-truth
 1255 correctness alone, leaving the verbalized confidence channel uninformative for L0 NoMemory.
 1256 Example: GPT-5.4 above (5/5 turns at conf ≥ 0.94).
- 1257 3. **Parser failure on free-form output.** A small fraction of agent responses do not match the regex
 1258 Answer: [A-D] (e.g., reasoning models that emit a long chain of thought before finally an-
 1259 swering). These are recorded as agent_answer = None and counted wrong; the user simulator
 1260 still reacts to the (truncated) output, so the turn contributes uninformative feedback. Example:
 1261 Qwen-Plus T5.
- 1262 4. **Decoder UNCLEAR on follow-up questions.** The persona’s “Could you elaborate on...?”
 1263 pattern, especially in academic style, is decoded as UNCLEAR more often than CORRECT.
 1264 Algorithmic L2 estimators that downweight UNCLEAR turns therefore see fewer effective
 1265 samples on these personas, slowing per-domain convergence.
- 1266 5. **Inflated zero-shot prior on Claude.** On all three Claude models the L1 ZS-SK estimate (0.460–
 1267 0.482 averaged across domains) is systematically below the L0 NoMemory anchor (0.557–0.603)
 1268 and below the true tier accuracy. Asking Claude to estimate its per-domain accuracy zero-shot
 1269 returns a conservative figure that under-estimates strong-tier domains specifically; the body’s H2
 1270 family-split observation is the aggregate consequence of this failure mode.

1271 I.3 Prompt templates

1272 The five prompt templates below are reproduced verbatim from the released code
 1273 (scripts/capbound_baselines_v4.py and scripts/classic_baselines_v4.py).

1274 **Agent answer prompt.** Used at every turn to elicit an answer and a confidence value.

1275 Answer the question and give your confidence 0-100%.
 1276
 1277 Question: {question}
 1278 Choices:
 1279 {choices_one_per_line}
 1280
 1281 Format your response exactly as:
 1282 Answer: [A/B/C/D]
 1283 Confidence: [0-100]

1284 **Simulated user prompt.** The cross-family LLM that plays the persona receives this template at
 1285 every turn. The scenario variable is sampled from a per-persona distribution that controls the
 1286 in/out-of-expertise reliability.

1287 Your previous question: {question}
 1288
 1289 The assistant’s answer: {agent_answer}
 1290
 1291 [Stage direction - do NOT quote]
 1292 {scenario}
 1293
 1294 Now write your natural next response (1-2 sentences):

1295 **Feedback decoder prompt.** A separate LLM call classifies the persona’s free-form response into a
 1296 structured label.

1297 A user is responding to an AI assistant’s answer. Classify whether
 1298 the user’s response implies the assistant’s answer was CORRECT,
 1299 WRONG, or UNCLEAR.
 1300
 1301 User’s response: "{user_text}"
 1302
 1303 Rules:
 1304 - CORRECT: user explicitly confirms (e.g., "That’s right",
 1305 "Exactly")

1306 - **WRONG:** user explicitly corrects or contradicts
1307 (e.g., "Actually...", "I'm afraid that's not right")
1308 - **UNCLEAR:** user asks follow-up, thanks without confirming,
1309 expresses vague uncertainty, or moves on to unrelated
1310 topic
1311
1312 Reply with exactly one word: **CORRECT**, **WRONG**, or **UNCLEAR**.

1313 **L1 ZS-SK self-assessment prompt.** Asked once per domain at the end of the session, with no
1314 interaction history.

1315 You are reflecting on your own performance in the domain:
1316 "{domain}".
1317 Below is a summary of your interaction history in this domain:
1318
1319 {context}
1320
1321 Based on this information, estimate what percentage of questions
1322 you answered correctly in this domain.
1323 Reply with **ONLY** a single integer between 0 and 100.

1324 **L2 LSA in-context aggregation prompt.** Asked at every turn before the agent answers, condition-
1325 ing on the recent history of (q, a, c, f) tuples.

1326 Based on the following recent interaction history with the user,
1327 estimate your likely accuracy on the next question.
1328
1329 Recent history:
1330 {history_text}
1331
1332 The next question is in domain: {domain}
1333 Next question: {question}
1334
1335 Based on how the user reacted to your previous answers in similar
1336 domains, what is your estimated probability of answering correctly?
1337 Reply with **ONLY** a number between 0 and 100, nothing else.

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1341 paper's contributions and scope?
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1344 ladder, the CBF metric, and the empirical findings (H1 ruled out, H2 family-split, H3 located
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1672 Question: Does the paper describe the usage of LLMs if it is an important, original, or
1673 non-standard component of the core methods in this research? Note that if the LLM is used
1674 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
1675 scientific rigor, or originality of the research, declaration is not required.

1676 Answer: [\[Yes\]](#)

1677 Justification: LLMs are simultaneously the subject of the benchmark and components of its
1678 protocol: as agents under evaluation, as the cross-family simulated user, and as the feedback
1679 decoder. Full configuration is documented in §3 and Appendix G.3.

1680 Guidelines:

- 1681 • The answer [\[N/A\]](#) means that the core method development in this research does not
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1684 be described.